HV10K
OFF-GRID INVERTER

User Manual V1.0

IMPORTANT SAFETY INSTRUCTIONS
SAVE THESE INSTRUCTIONS

The contents of this manual are subject to change without notice. Please obtain the latest version from BLUETTI official website:

https://ng.bluettipower.com/pages/user-guides

If you have any questions or concerns about this manual, please contact BLUETTI support for further assistance.

BLUETTI

Table Of Contents

ABOUT THIS MANUAL

Purpose

This manual describes the assembly, installation, operation and troubleshooting of this unit. Please read this manual carefully before installations and operations. Keep this manual for future reference.

Scope

This manual provides safety and installation guidelines as well as information on tools and wiring.

SAFETY INSTRUCTIONS

WARNING: This chapter contains important safety and operating instructions. Read and keep this manual for future reference.

1. Before using the unit, read all instructions and cautionary markings on the unit, the batteries and all appropriate sections of this manual.
2. **CAUTION** --To reduce risk of injury, charge only deep-cycle lead acid type rechargeable batteries. If any other types of batteries are used, follow the manufacturer's instructions carefully.
3. Do not disassemble the unit. Take it to a qualified service center when service or repair is required. Incorrect re-assembly may result in a risk of electric shock or fire.
4. To reduce risk of electric shock, disconnect all wirings before attempting any maintenance or cleaning. Turning off the unit will not reduce this risk.
5. **CAUTION** – Only qualified personnel can install this device with battery.
6. **NEVER** charge a frozen battery.
7. For optimum operation of this inverter/charger, please follow required spec to select appropriate cable size. It's very important to correctly operate this inverter/charger.
8. Be very cautious when working with metal tools on or around batteries. A potential risk exists to drop a tool to spark or short circuit batteries or other electrical parts and could cause an explosion.
9. Please strictly follow installation procedure when you want to disconnect AC or DC terminals. Please refer to INSTALLATION section of this manual for the details.
10. One piece of 150A fuse is provided as over-current protection for the battery supply.
11. GROUNDING INSTRUCTIONS -This inverter/charger should be connected to a permanent grounded wiring system. Be sure to comply with local requirements and regulation to install this inverter.
12. NEVER cause AC output and DC input short circuited. Do NOT connect to the mains when DC input short circuits.
13. **Warning!!** Only qualified service persons are able to service this device. If errors still persist after following troubleshooting table, please send this inverter/charger back to local dealer or service center for maintenance.
14. **WARNING:** Because this inverter is non-isolated, only three types of PV modules are acceptable: single crystalline, poly crystalline with class A-rated and CIGS modules. To avoid any malfunction, do not connect any PV modules with possible current leakage to the inverter. For example, grounded PV modules will cause current leakage to the inverter. When using CIGS modules, please be sure NO grounding.
15. **CAUTION:** It's requested to use PV junction box with surge protection. Otherwise, it will cause damage on inverter when lightning occurs on PV modules.
16. After the machine is powered off, internal capacitors may still retain electrical charge. The capacitors require up to 60 seconds to fully discharge. It is recommended to wait until the capacitors are completely discharged before performing any operation.

17. If the equipment is not used in accordance with the manufacturer's instructions, the protective measures provided by the equipment may be impaired.

18. Warning: Once exposed to light, the photovoltaic (PV) array will generate DC voltage and supply it to the PCE.

INTRODUCTION

This is a multi-function inverter/charger, combining functions of inverter, solar charger and battery charger to offer uninterruptible power support with portable size. Its comprehensive LCD display offers user-configurable and easy-accessible button operation such as battery charging current, AC/solar charger priority, and acceptable input voltage based on different applications.

Features

- Pure sine wave inverter
- Wide PV input range
- Built-in BMS communication port
- Built-in anti-dust kit
- Inverter running without battery
- Configurable input voltage range for home appliances and personal computers via LCD setting
- Configurable battery charging current based on applications via LCD setting
- Configurable AC/Solar Charger priority via LCD setting
- Compatible to mains voltage or generator power
- Overload/ Over temperature/ short circuit protection
- Smart battery charger design for optimized battery performance

Basic System Architecture

The following illustration shows basic application for this inverter/charger. It also includes following devices to have a complete running system:

- Generator or Utility.
- PV modules

Consult with your system integrator for other possible system architectures depending on your requirements.

This inverter can power all kinds of appliances in home or office environment, including motor-type appliances such as tube light, fan, refrigerator and air conditioner.

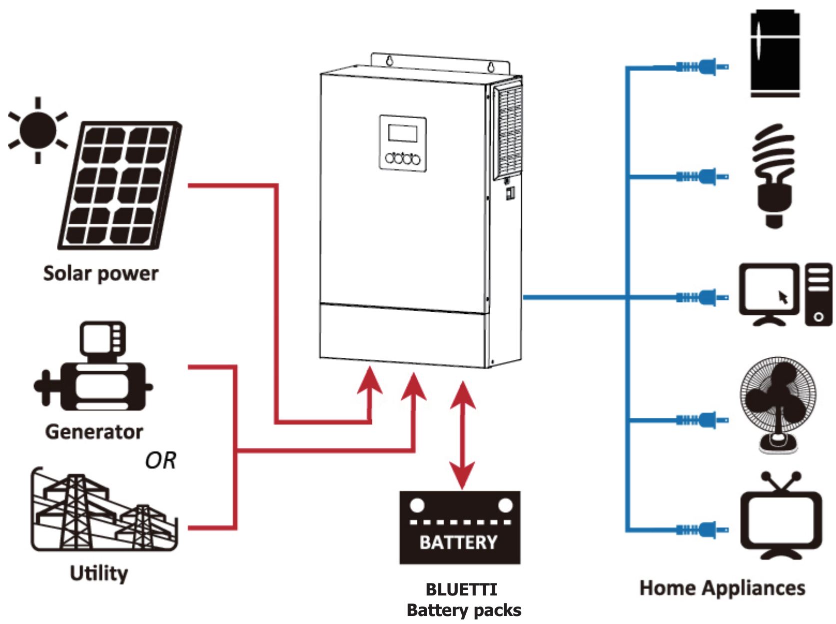
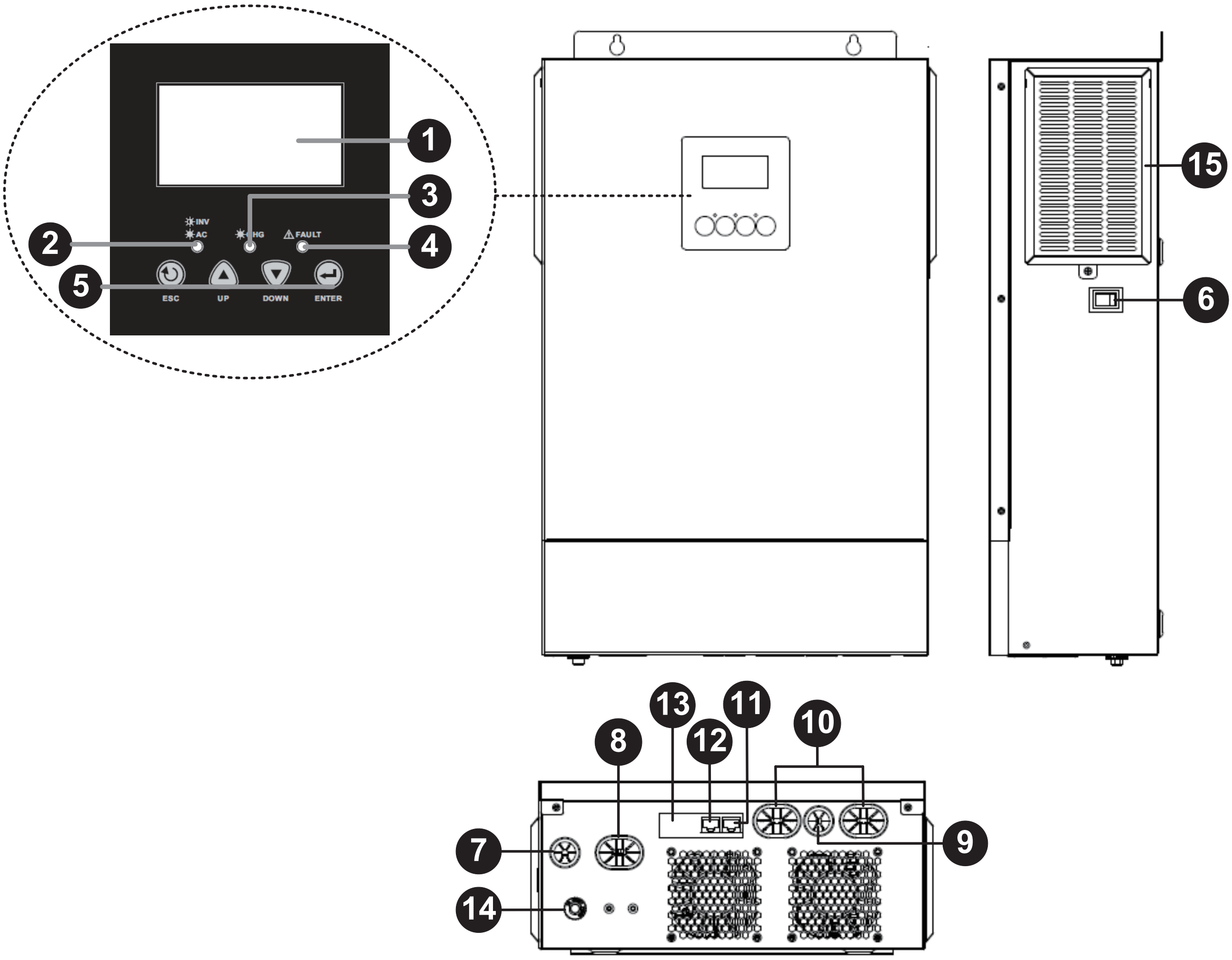


Figure 1 Solar Power System

Product Overview



1. LCD display
2. Status indicator
3. Charging indicator
4. Fault indicator
5. Function keys
6. Power on/off switch
7. AC input
8. AC output
9. PV input
10. Battery input
11. RS-232 communication port
12. BMS communication port
13. Internal WiFi
14. Input circuit breaker
15. Anti-dust filter

INSTALLATION

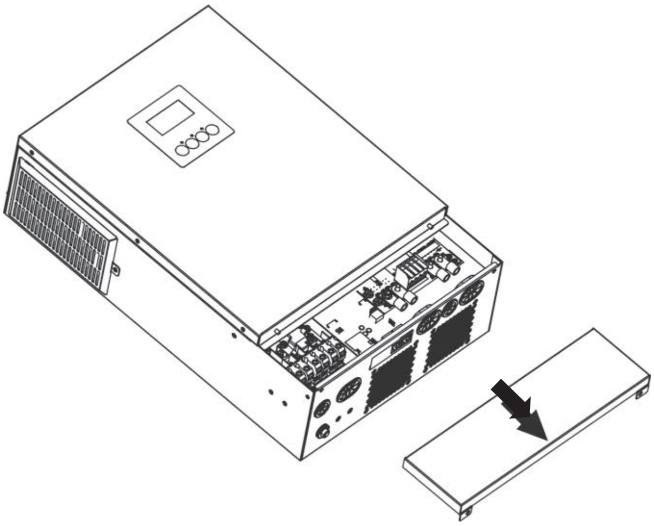
Unpacking and Inspection

Before installation, please inspect the unit. Be sure that nothing inside the package is damaged. You should have received the following items inside of package:

- The unit x 1
- User manual x 1
- Communication cable x 1
- DC Fuse x 2

Preparation

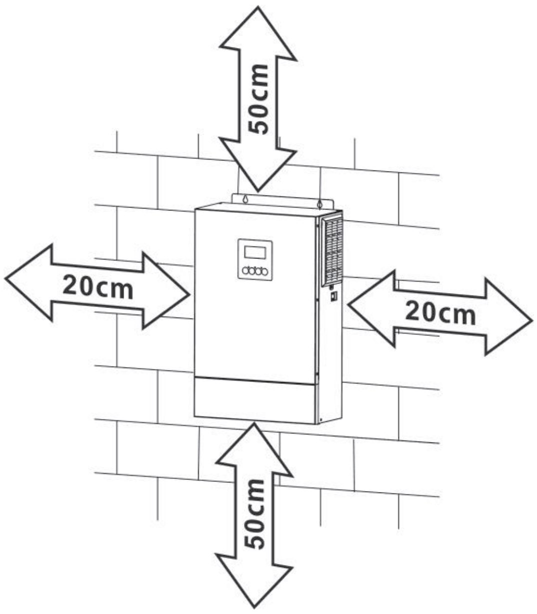
Before connecting all wirings, please take off bottom cover by removing two screws as shown below.



Mounting the Unit

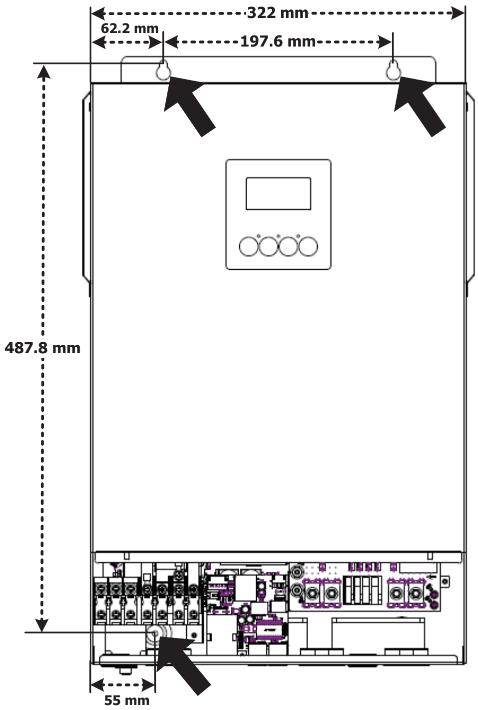
Consider the following points before selecting where to install:

- Do not mount the inverter on flammable construction materials.
- Mount on a solid surface
- Install this inverter at eye level in order to allow the LCD display to be read at all times.
- For proper air circulation to dissipate heat, allow a clearance of approx. 20 cm to the side and approx. 50 cm above and below the unit.
- The ambient temperature should be between 0°C and 55°C to ensure optimal operation.
- The recommended installation position is to be adhered to the wall vertically.
- Be sure to keep other objects and surfaces as shown in the diagram to guarantee sufficient heat dissipation and to have enough space for removing wires.



⚠ **SUITABLE FOR MOUNTING ON CONCRETE OR OTHER NON-COMBUSTIBLE SURFACE ONLY.**

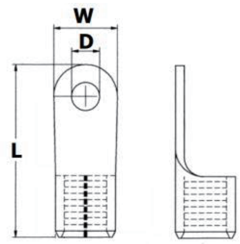
Drill three holes in the marked location and then install the unit by screwing three screws. It's recommended to use M4 or M5 screws.



Battery Connection

CAUTION: For safety operation and regulation compliance, it's requested to install a separate DC over-current protector or disconnect device between battery and inverter. It may not be requested to have a disconnect device in some applications, however, it's still requested to have over-current protection installed. Please refer to typical amperage in below table as required fuse or breaker size.

Ring Terminal:



WARNING! All wiring must be performed by a qualified personnel.
WARNING! It's very important for system safety and efficient operation to use appropriate cable for battery connection. To reduce risk of injury, please use the proper recommended cable as below.

Recommended battery cable size:

Model	Typical Amperage	Wire Size	Cable (mm^2)	Dimensions for Ring Terminal			Torque value (max)
				W (mm)	D (mm)	L (mm)	
HV10K	230A	2 x 2AWG	38	13.2	6.4	37.6	3 Nm

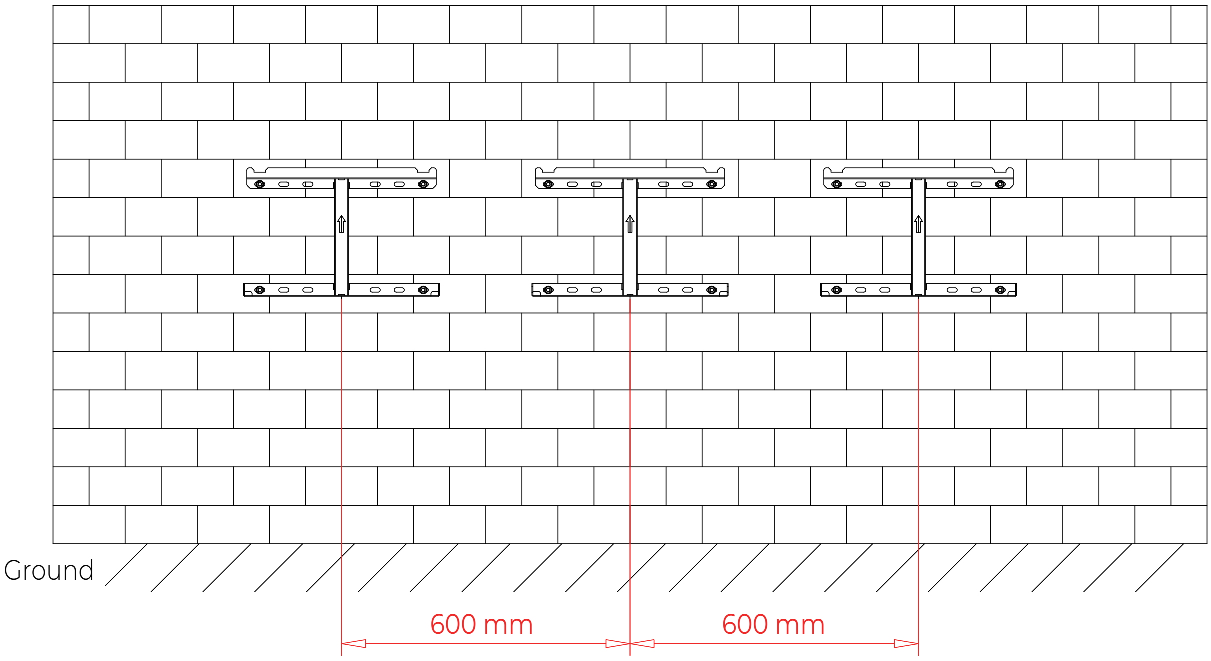
⚠	**WARNING: Shock Hazard** Power must be disconnected before connecting to the battery.
⚠	**CAUTION!!** Do not place anything between inverter terminals and the ring terminals. Otherwise, overheating may occur. **CAUTION!!** Do not apply anti-oxidant substance on the terminals before terminals are securely tightened. **CAUTION!!** Before making the final DC connection or closing DC breaker/disconnector, be sure positive (+) must be connected to positive (+) and negative (-) must be connected to negative (-).

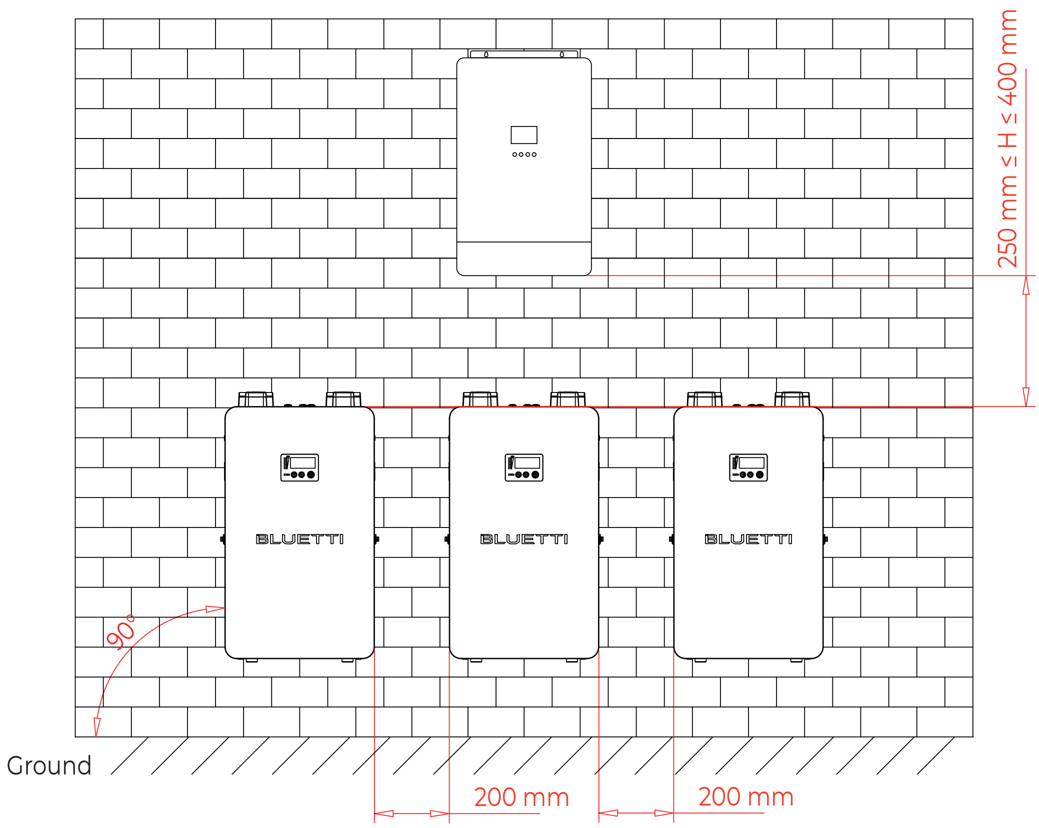
Connect HB500 in Parallel

Notes:
- Connect HB500 units in parallel only, and always keep them upright.
- If the HB500 is over-discharged, disconnect all loads immediately.

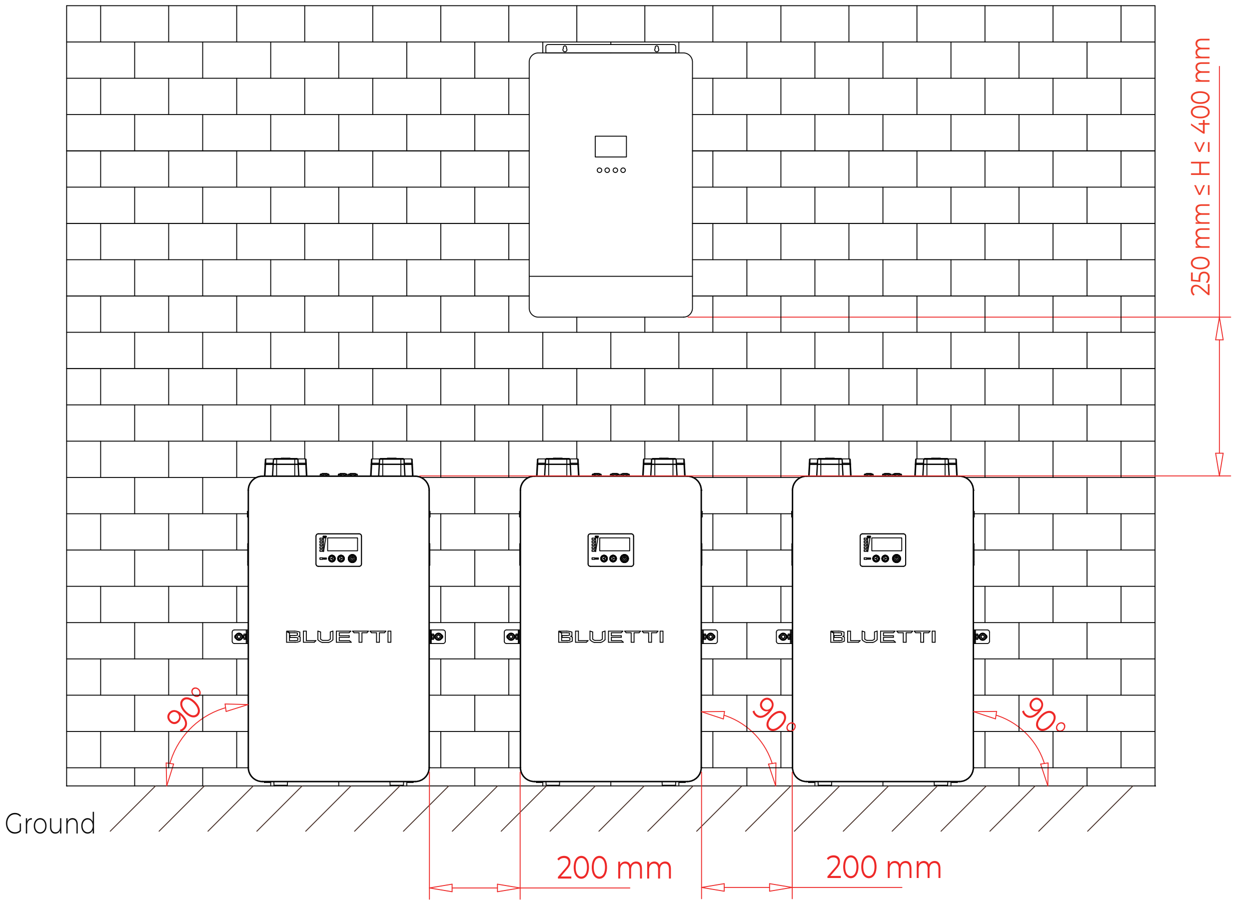
1: Securely mount all HB500 units, leaving enough clearance for neat cable routing, as shown below.

Wall-mounted:

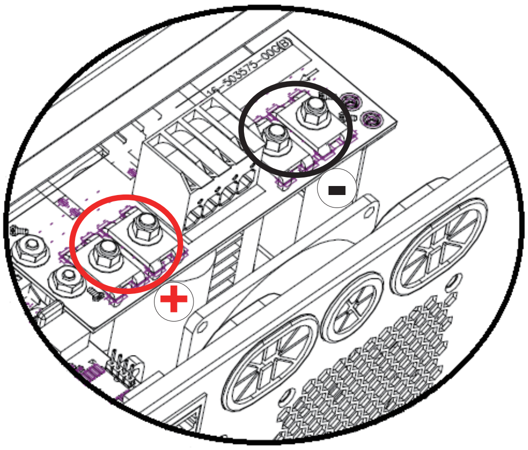




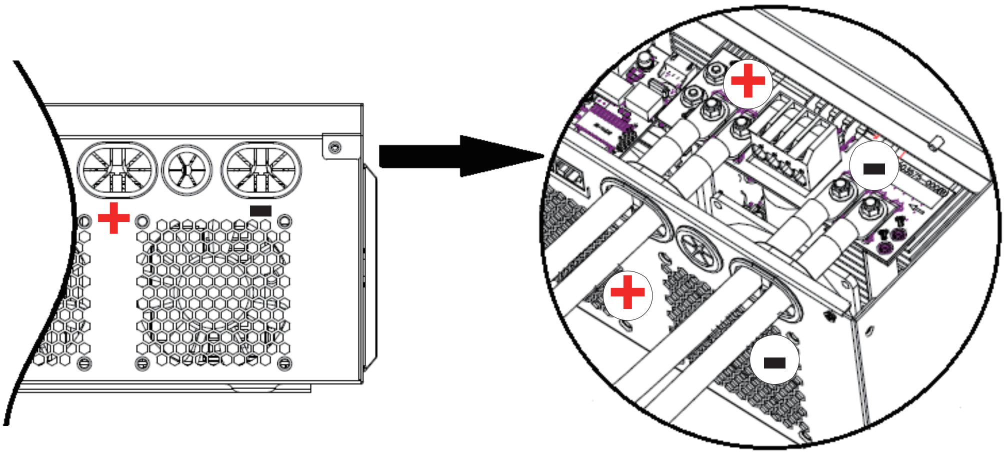
Floor-mounted:



2. Remove the nuts on the battery terminals of the inverter.



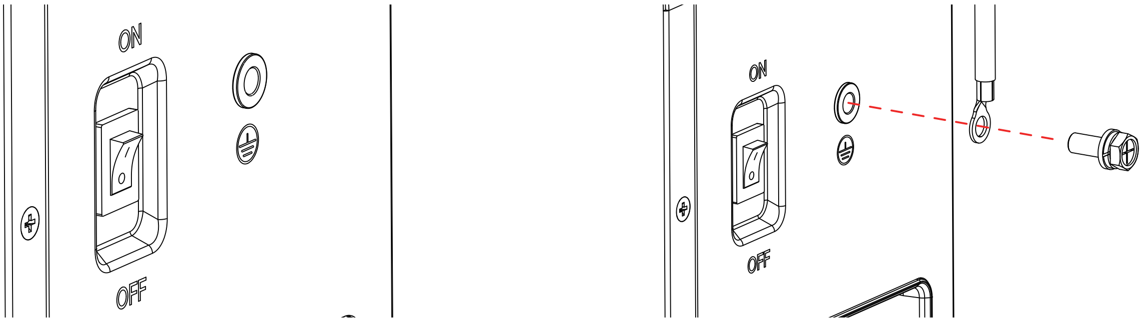
3. Insert the ring terminals of battery cable flatly into battery terminals. Then, make sure the nuts are screwed firmly with correct torque.



4: Turn the Emergency Stop Switch **OFF**. The display will go blank. Make sure all switches on the inverter and distribution panel are **OFF**. Before installation, please inspect the unit. Be sure that nothin

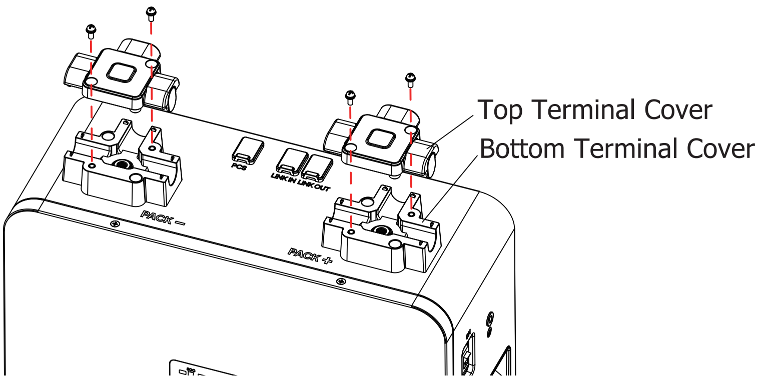
5: Connect one end of the grounding cable to the unit's grounding terminals using an M6×14 screw.

Recommended torque: 30 kgf·cm



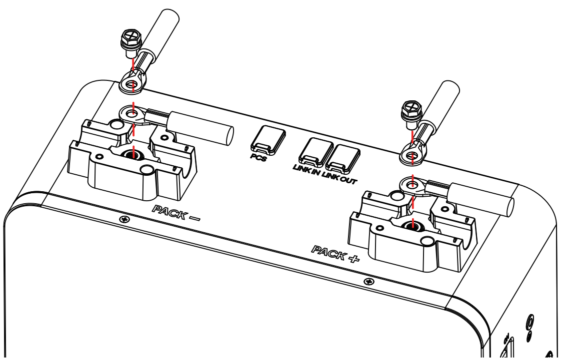
6: Connect the other end of the cable to a reliable grounding point.

7: Unscrew the top terminal covers.

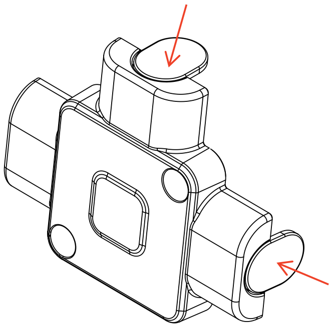


8: Tightly secure the red power cable to the positive terminal and the black power cable to the negative terminal with M8×14 screws.

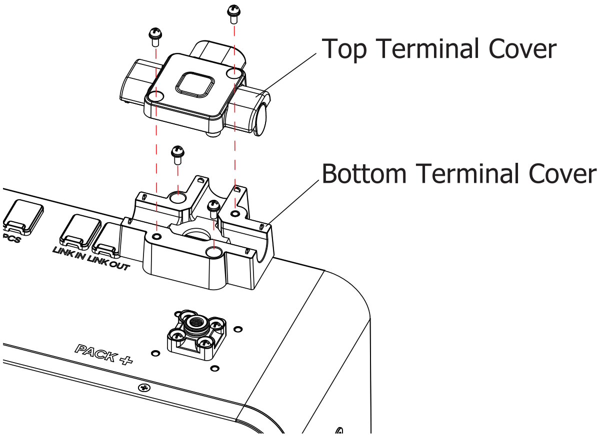
Recommended torque: 80 kgf·cm



9: Remove the tabs on the covers if needed for cable routing.



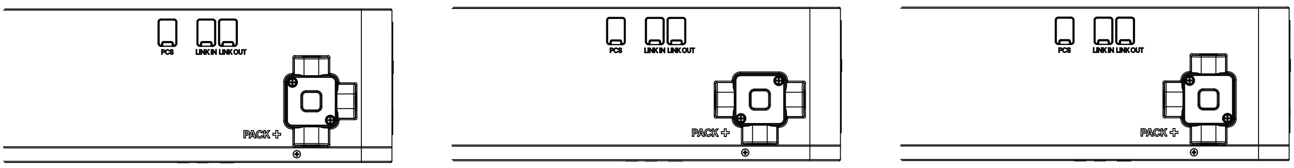
10: To adjust the cable exit direction, remove the bottom parts of the terminal covers.



11: Rotate the covers 90° or 180°, then reinstall the bottom and top covers.

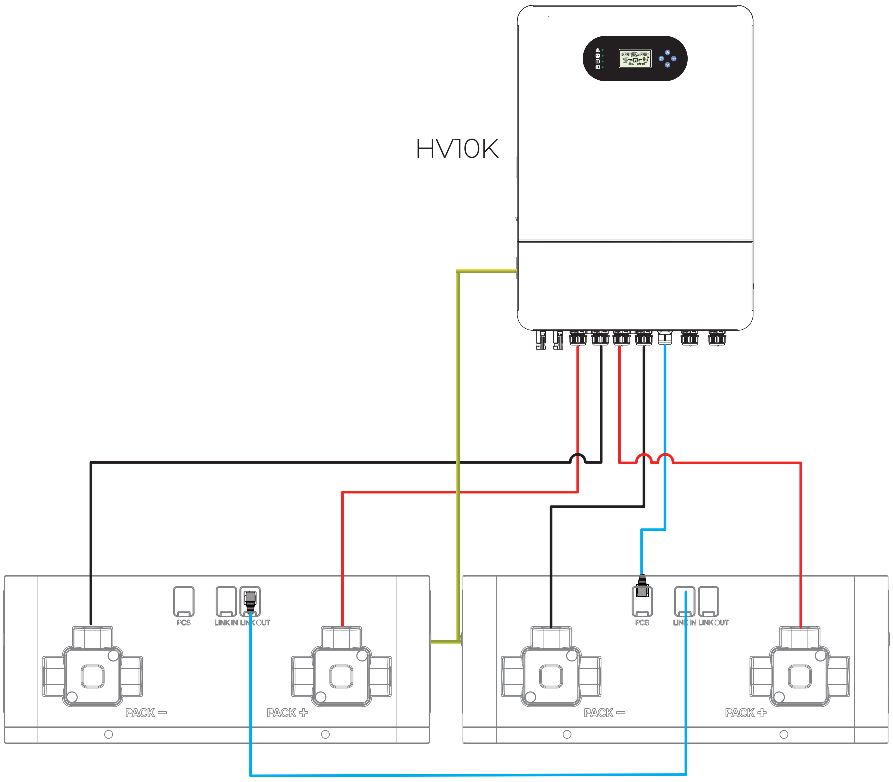
Recommended torque: 6 kgf·cm

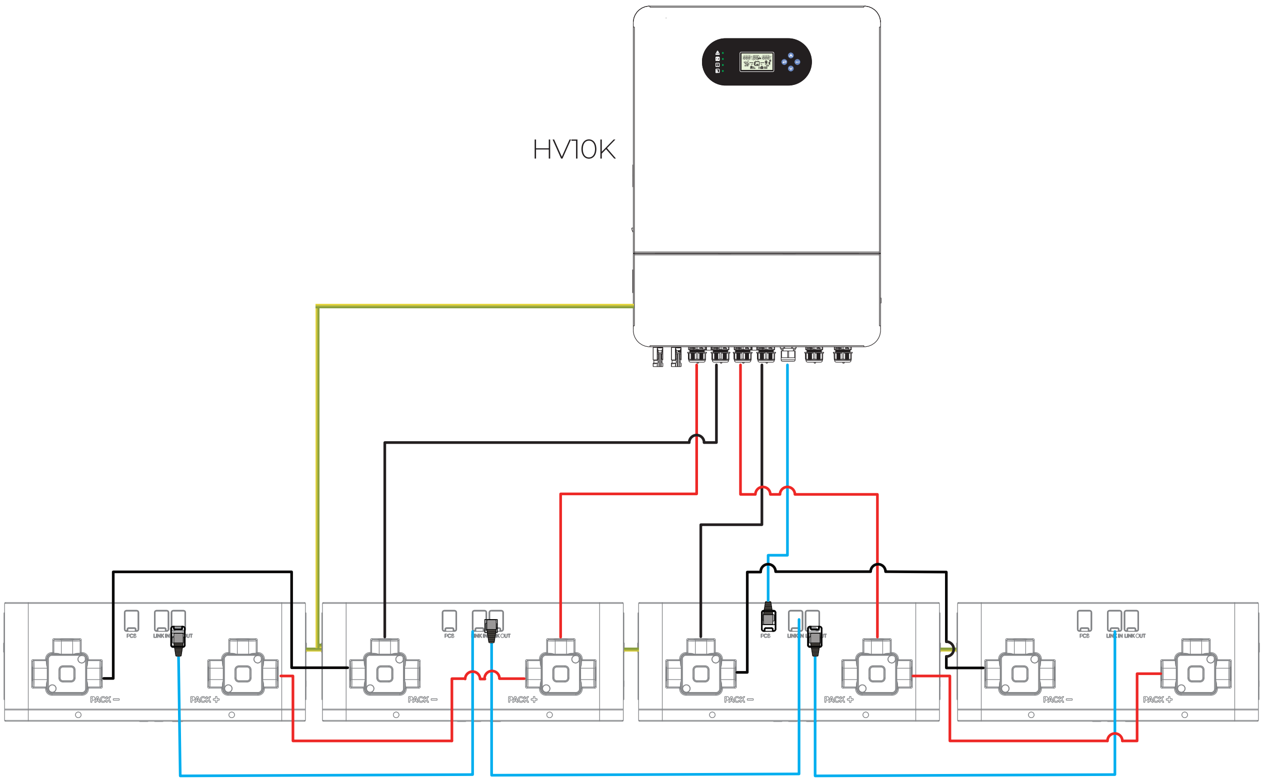
Note: Connect up to two cables to each terminal.

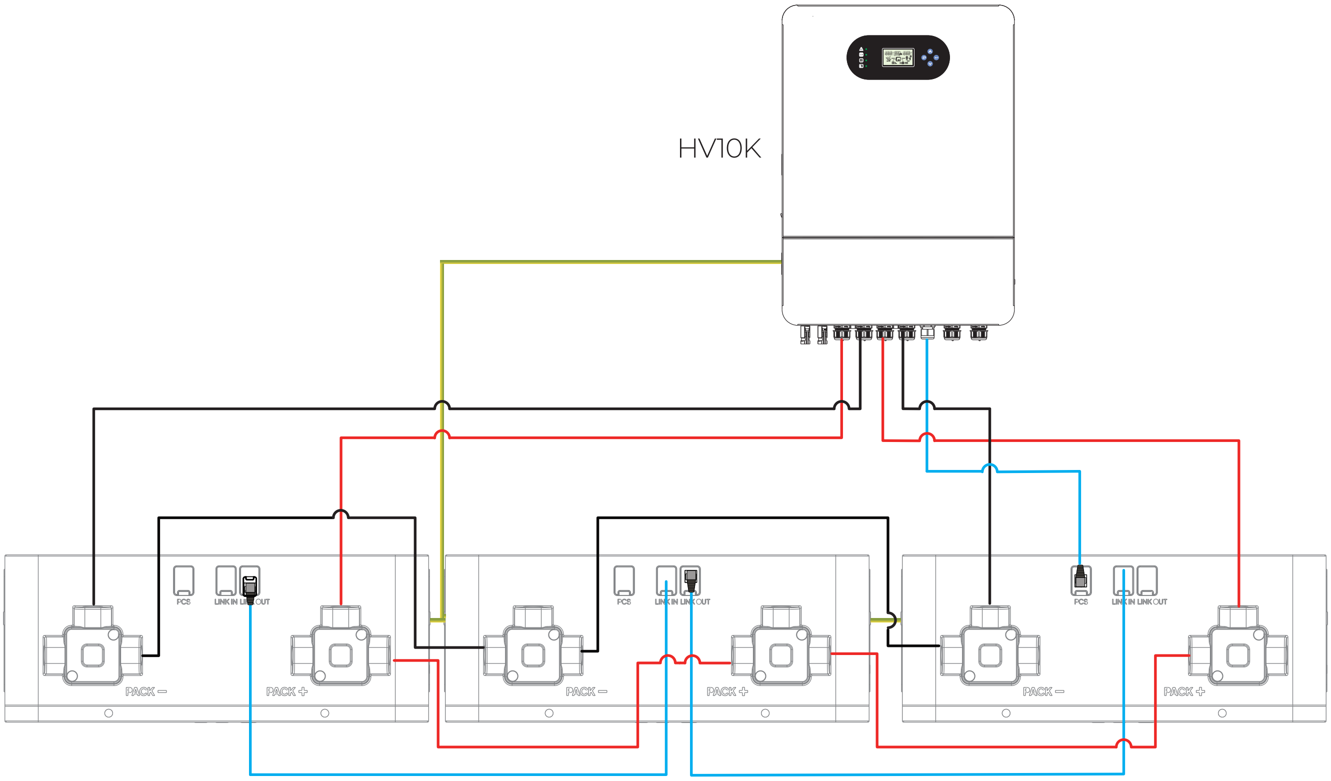


Connect multiple HB500 units in parallel as shown.

Inverter with Dual DC Inputs







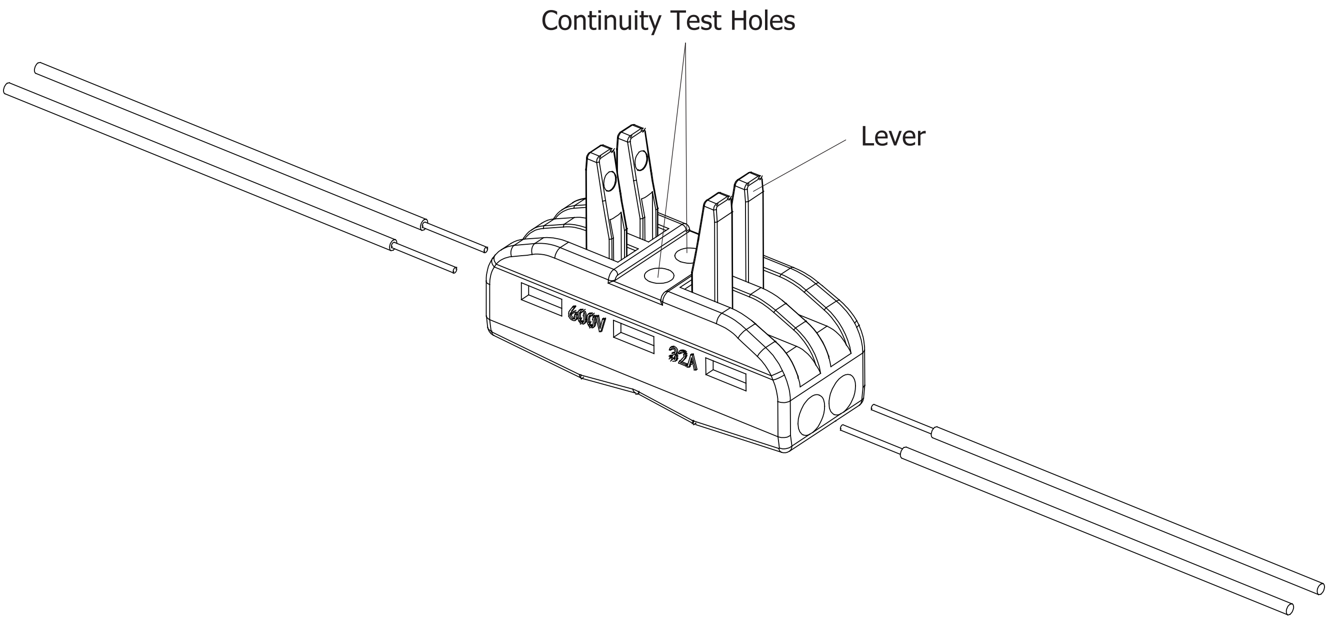
HV10K
PCS
LINK IN
LINK OUT
PACK −
PACK +
PCS
LINK IN
LINK OUT
PACK −
PACK +
PCS
LINK IN
LINK OUT
PACK −
PACK +

Rewire the Communication Cable

When connecting to a third-party inverter, rewire the communication cable if needed:

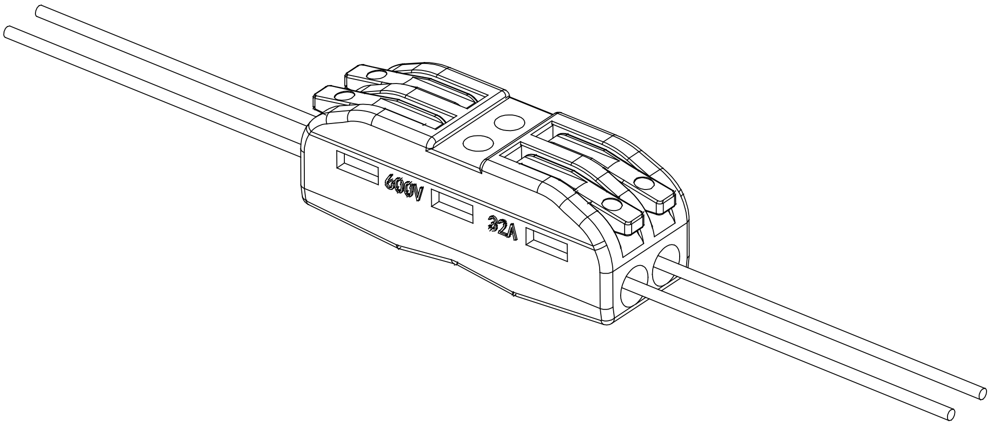
1: Cut the Ethernet cable. Check the pin definitions for the HB500 and the inverter communication ports.

Select the required wires and strip 9–12 mm of insulation from each.



2: Open the levers, insert the stripped wires, and press the levers down to secure them.

3: Test for continuity to confirm the rewiring is complete.



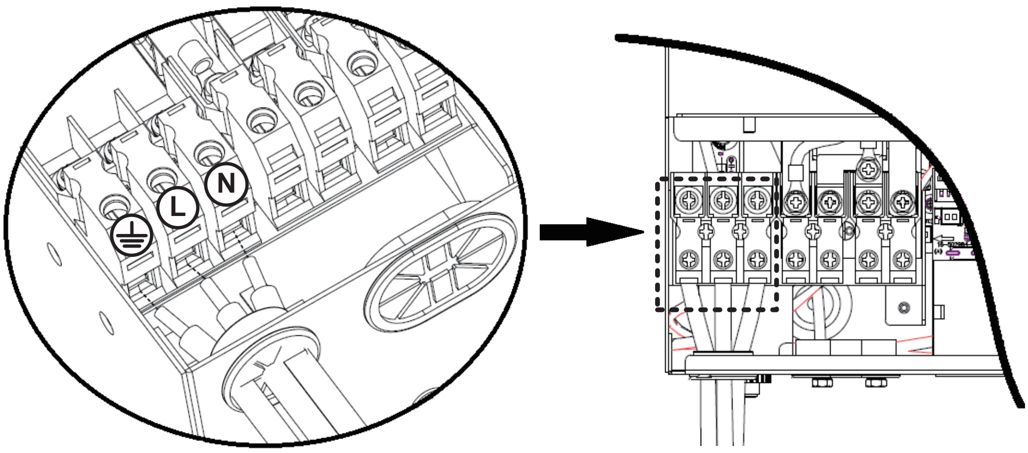
AC Input/Output Connection

CAUTION!! Before connecting to AC input power source, please install a **separate** AC breaker between inverter and AC input power source. This will ensure the inverter can be securely disconnected during maintenance and fully protected from over current of AC input. The recommended spec of AC breaker is 20A.
CAUTION!! There are two terminal blocks with "IN" and "OUT" markings. Please do NOT mis-connect input and output connectors.

WARNING! All wiring must be performed by a qualified personnel.
WARNING! It's very important for system safety and efficient operation to use appropriate cable for AC input connection. To reduce risk of injury, please use the proper recommended cable size as below.
Suggested cable requirement for AC wires

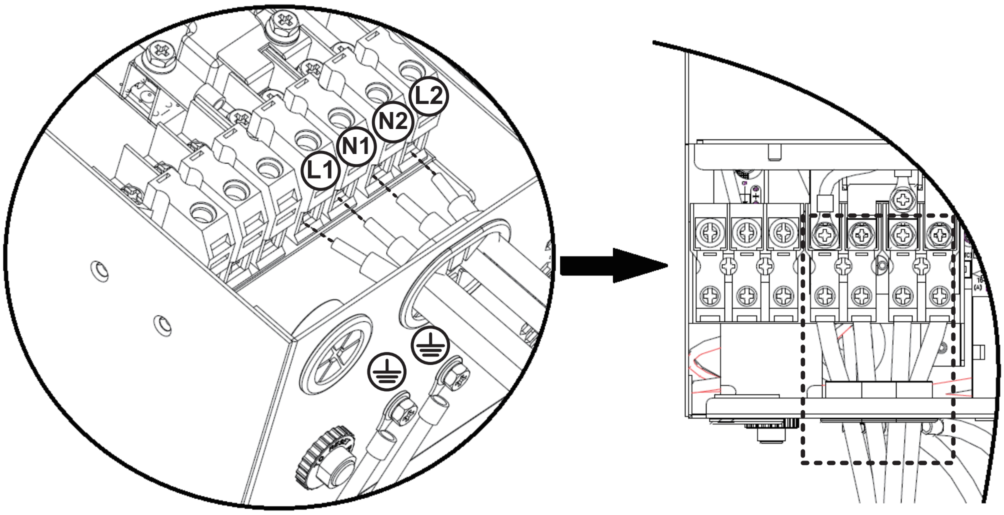
Model	Gauge	Cable (mm^2)	Torque Value
HV10K	8 AWG	8	1.4~1.6 Nm

Please follow below steps to implement AC input/output connection:
1. Before making AC input/output connection, be sure to open DC protector or disconnector first.
2. Remove insulation sleeve 10mm for seven conductors. And shorten phase L and neutral conductor N 3 mm.
3. Insert AC input wires according to polarities indicated on terminal block and tighten the terminal screws. Be sure to connect PE protective conductor (⏚) first.
   ⏚→**Ground (yellow-green)**
   L→LINE (brown or black)
   N→Neutral (blue)



⚠	**WARNING:** Be sure that AC power source is disconnected before attempting to hardwire it to the unit.

4. This inverter is equipped with dual-output. There are four terminals (L1/N1, L2/N2) available on output port. It's set up through LCD program or monitoring software to turn on and off the second output. Refer to "LCD setting" section for the details.
   Insert AC output wires according to polarities indicated on terminal block and tighten terminal screws. Be sure to connect PE protective conductor (⏚) first.
   ⏚→**Ground (yellow-green)**
   L1→LINE (brown or black)
   N1→Neutral (blue)
   L2→LINE (brown or black)
   N2→Neutral (blue)



5. Make sure the wires are securely connected.

CAUTION: Appliances such as air conditioner are required at least 2~3 minutes to restart because it's required to have enough time to balance refrigerant gas inside of circuits. If a power shortage occurs and recovers in a short time, it will cause damage to your connected appliances. To prevent this kind of damage, please check manufacturer of air conditioner if it's equipped with time-delay function before installation. Otherwise, this inverter/charger will trig overload fault and cut off output to protect your appliance but sometimes it still causes internal damage to the air conditioner.

PV Connection

CAUTION: Before connecting to PV modules, please install **separately** a DC circuit breaker between inverter and PV modules.

WARNING! It's very important for system safety and efficient operation to use appropriate cable for PV module connection. To reduce risk of injury, please use the proper recommended cable size as below.

Wire Size	Cable (mm^2)
10~12 AWG	4~6

WARNING: Because this inverter is non-isolated, only three types of PV modules are acceptable: single crystalline, poly crystalline with class A-rated and CIGS modules.
To avoid any malfunction, do not connect any PV modules with possible current leakage to the inverter. For example, grounded PV modules will cause current leakage to the inverter. When using CIGS modules, please be sure NO grounding.
CAUTION: It's requested to use PV junction box with surge protection. Otherwise, it will cause damage on inverter when lightning occurs on PV modules.
Never directly touch the terminals of inverter. It might cause lethal electric shock.

PV Module Selection:
When selecting proper PV modules, please be sure to consider below parameters:
1. Open circuit Voltage (Voc) of PV modules not exceeds max. PV array open circuit voltage of inverter.
2. Open circuit Voltage (Voc) of PV modules should be higher than min. battery voltage.

INVERTER MODEL	HV10K	
Max. PV Array Power	7000W x 2	
Max. PV Array Open Circuit Voltage	500Vdc	
PV Array MPPT Voltage Range	90~450Vdc	
Max. PV input Current	22A x 2	27A x 2

This unit is applied with two strings of PV array. Please make sure that the maximum current load for each string cannot beyond of the max PV input current spec.

Take 600Wp PV module as an example, the recommended configurations are listed as below table.

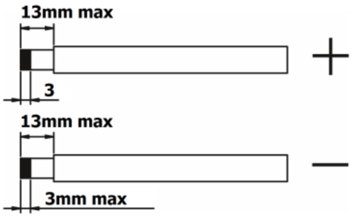
	Solar Input 1	Solar Input 2	Q'ty of panels	Total input power
	Min in series: 3pcs, per input, Max. in series: 9pcs, per input			
Solar Panel Spec. (reference) - 600 - Vmp: 45.30Vdc - Imp: 13.25A - Voc: 53.50Vdc - Isc: 14.03A	3pcs in series	x	3pcs	1800W
	x	3pcs in series	3pcs	1800W
	9pcs in series	x	9pcs	5400W
	9pcs in series	9pcs in series	18pcs	10800W
	5pcs in series, 2 Strings in parallel	5pcs in series, 2 Strings in parallel	20pcs	12000W
	6pcs in series, 2 Strings in parallel	6pcs in series, 2 Strings in parallel	24pcs	14400W

Take 660Wp PV module as an example, the recommended configurations are listed as below table.

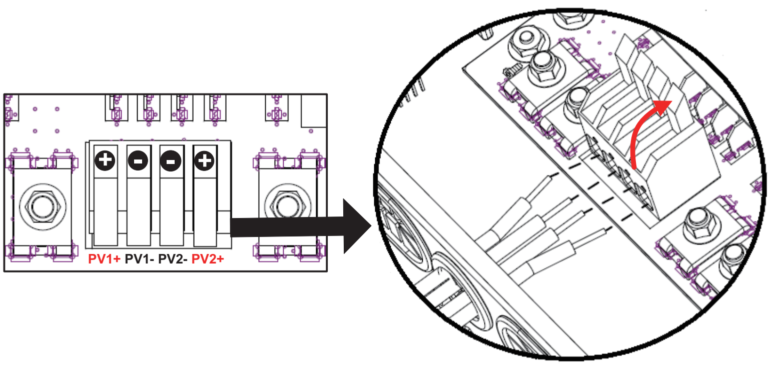
	Solar Input 1	Solar Input 2	Q'ty of panels	Total input power
	Min in series: 3pcs, per input Max. in series: 11pcs, per input for HV10K model			
Solar Panel Spec. (reference) - 660Wp - Vmp: 37.80Vdc - Imp: 17.46A - Voc: 45.60Vdc - Isc: 18.55A	3pcs in series	x	3pcs	1980W
	x	3pcs in series	3pcs	1980W
	9pcs in series	x	9pcs	5940W
	10pcs in series	x	10pcs	6600W
	9pcs in series	9pcs in series	18pcs	11880W
	10pcs in series	10pcs in series	20pcs	13200W
	11pcs in series	11pcs in series	22pcs	14520W

PV Module Wire Connection

Please follow below steps to implement PV module connection:

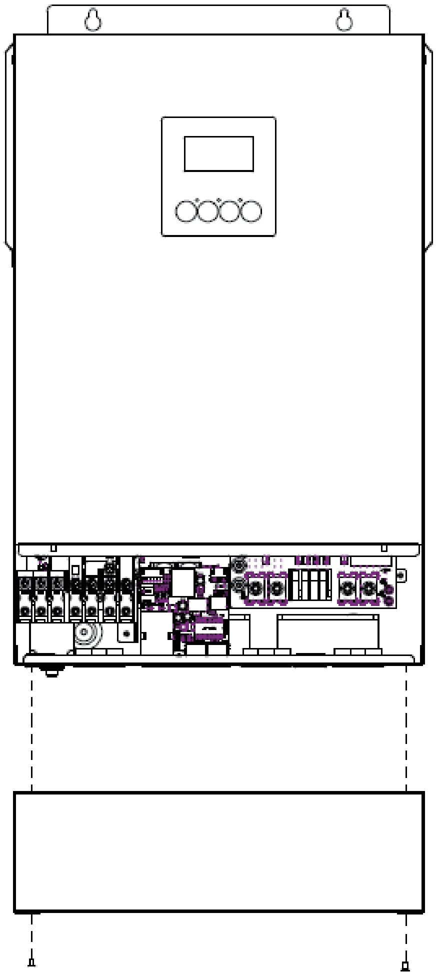


1. Remove insulation sleeve 11~13 mm for positive and negative conductors.
2. Suggest to put bootlace ferrules on the end of positive and negative wires with a proper crimping tool.
3. Check correct polarity of wire connection from PV modules and PV input terminals. Connect positive pole (+) of connection wire to positive pole (+) of PV input terminals, and connect negative pole (-) of connection wire to negative pole (-) of PV input terminals. Pull the spring cover upward to loosen the PV terminal block, insert the wire conductors into it, and then press down on the spring cover. Make sure the wires are securely locked on PV terminals.



Final Assembly

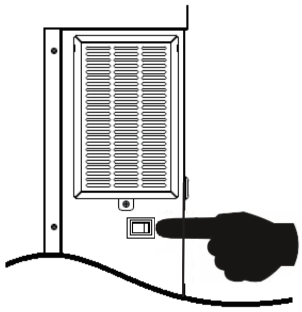
After connecting all wirings, please put bottom cover back by screwing screws as shown below.



OPERATION

Power ON/OFF

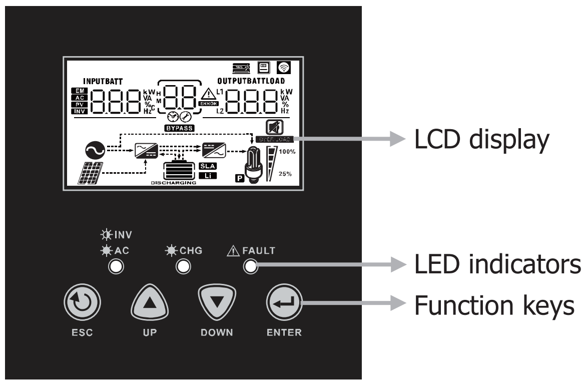
Side view of unit



Once the unit has been properly installed and the batteries are connected well, simply press On/Off switch to turn on the unit.

Operation and Display Panel

The operation and display panel, shown in below chart, is on the front panel of the inverter. It includes three indicators, four function keys and a LCD display, indicating the operating status and input/output power information.



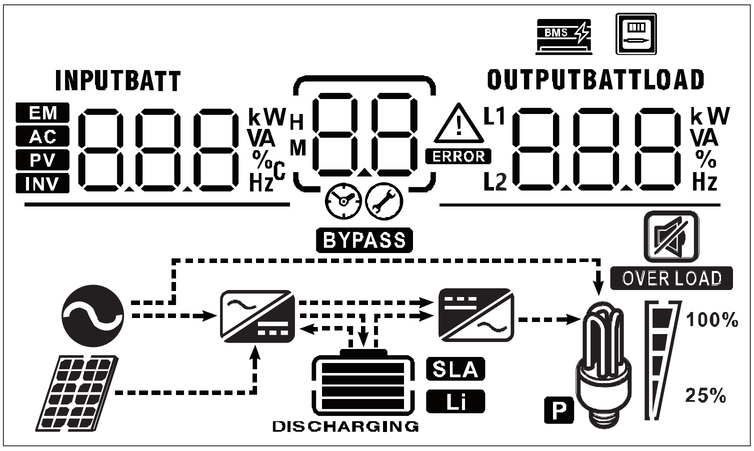
LED Indicator

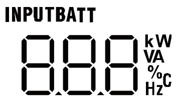
LED Indicator			Messages
INV AC	Green	Solid On	Output is powered by utility in Line mode.
		Flashing	Output is powered by battery or PV in battery mode.
CHG	Green	Solid On	Battery is fully charged.
		Flashing	Battery is charging.
FAULT	Red	Solid On	Fault occurs in the inverter.
		Flashing	Warning condition occurs in the inverter.

Function Keys

Function Key		Description
	ESC	To exit setting mode
▲	UP	To go to previous selection
▼	DOWN	To go to next selection
	ENTER	To confirm the selection in setting mode or enter setting mode

LCD Display Icons



Icon	Function description
Input Source Information	
AC	Indicates the AC input.
PV1	Indicates the PV input information of Tracker 1
PV2	Indicates the PV input information of Tracker 2
	Indicate input voltage, input frequency, PV voltage, PV current, PV power, charger current, charger power, discharger current and battery voltage.
Configuration Program and Fault Information	
	Indicates the setting programs.
	Indicates the warning and fault codes. Warning:  flashing with warning code. Fault:  lighting with fault code.
Output Information	
	Indicate output voltage, output frequency, load percent, load in VA, load in Watt and discharging current.
Battery Information	
	Indicates battery level by 0-24%, 25-49%, 50-74% and 75-100%, charging /discharging status.

Battery in charging status, it will present battery charging status.

Status	Battery Capacity	LCD Display
Constant Current mode / Constant Voltage mode	0-24%	4 bars will flash in turns.
	25-49%	Bottom bar will be on and the other three bars will flash in turns.
	50-74%	Bottom two bars will be on and the other two bars will flash in turns.
	75-100%	Bottom three bars will be on and the top bar will flash.
Floating mode. Batteries are fully charged.		4 bars will be on.

Battery in discharging status, it will present battery capacity.

Battery Capacity	0-24%	25-49%	50-74%	75-100%
LCD Display				

Load Information

OVER LOAD	Indicates overload.
100% 25%	Indicates the load level by 0-24%, 25-49%, 50-74% and 75-100%.

0%~24%	25%~49%	50%~74%	75%~100%

Mode Operation Information

	Indicates unit connects to the mains.
	Indicates unit connects to the PV panel.
BYPASS	Indicates load is supplied by utility power.
	Indicates the utility charger circuit is working.
	Indicates the DC/AC inverter circuit is working.

Mute Operation

	Indicates unit alarm is disabled.

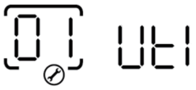
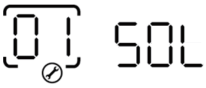
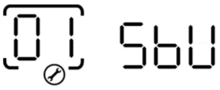
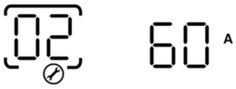
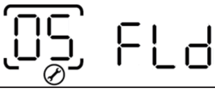
Other Information

BMS	Indicates BMS communication is established between the inverter and Lithium battery. It flashes while BMS is detected by inverter but communication can't be well established.
	Indicates the unit is connected to an external energy meter.

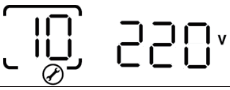
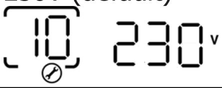
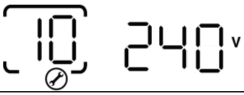
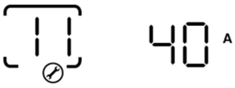
LCD Setting

After pressing and holding ENTER button for 3 seconds, the unit will enter setting mode. Press "UP" or "DOWN" button to select setting programs. And then, press "ENTER" button to confirm the selection or "ESC" button to exit.

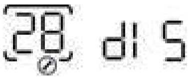
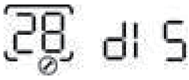
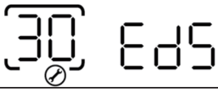
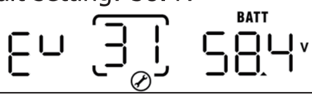
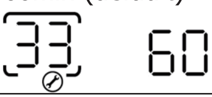
Setting Programs:

Program	Description	Selectable option	
00	Exit setting mode	Escape 	
01	Output source priority: To configure load power source priority	Utility first (default) 	Utility will provide power to the loads as first priority. Solar and battery energy will provide power to the loads only when utility power is not available.
		Solar first 	Solar energy provides power to the loads as first priority. If solar energy is not sufficient to power all connected loads, Utility energy will supply power to the loads at the same time.
		SBU priority 	Solar energy provides power to the loads as first priority. If solar energy is not sufficient to power all connected loads, battery energy will supply power to the loads at the same time. Utility provides power to the loads only when battery voltage drops to either low-level warning voltage or the setting point in program 12.
02	Maximum charging current: To configure total charging current for solar and utility chargers. (Max. charging current = utility charging current + solar charging current)	60A (default) 	Setting range is from 10A to 180A for HV10K model.
03	AC input voltage range	Appliances (default) 	If selected, acceptable AC input voltage range will be within 90-280VAC.
		UPS 	If selected, acceptable AC input voltage range will be within 170-280VAC.
05	Battery type	AGM (default) 	Flooded 

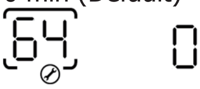
05	Battery type	User-Defined 05 USE	If "User-Defined" is selected, battery charge voltage and low DC cut-off voltage can be set up in program 26, 27 and 29.
		Pylontech battery 05 PYL	If selected, programs of 02, 26, 27 and 29 will be automatically set up. No need for further setting.
		BYD battery 05 bYd	If selected, programs of 02, 26, 27 and 29 will be automatically set up. No need for further setting.
		WECO battery 05 uEC	If selected, programs of 02, 12, 26, 27 and 29 will be auto-configured per battery supplier recommended. No need for further adjustment.
		Soltaro battery 05 SOL	If selected, programs of 02, 26, 27 and 29 will be automatically set up. No need for further setting.
		LIA-protocol compatible battery 05 LIA	Select "LIA" if using Lithium battery compatible to CAN protocol. If selected, programs of 02, 26, 27 and 29 will be automatically set up. No need for further setting.
		LIb-protocol compatible battery 05 LIb	Select "LIb" if using Lithium battery compatible to RS485 protocol. If selected, programs of 02, 26, 27 and 29 will be automatically set up. No need for further setting.
		3rd party Lithium battery 05 LIC	Select "LIC" if using Lithium battery not listed above. If selected, programs of 02, 26, 27 and 29 will be automatically set up. No need for further setting. Please contact the battery supplier for installation procedure.
06	Auto restart when overload occurs	Restart disable (default) 06 Ltd	Restart enable 06 LtE
07	Auto restart when over temperature occurs	Restart disable (default) 07 ttd	Restart enable 07 ttE
09	Output frequency	50Hz (default) 09 50 Hz	60Hz 09 60 Hz

10	Output voltage	220V 	230V (default) 
		240V 	
11	Maximum utility charging current Note: If setting value in program 02 is smaller than that in program in 11, the inverter will apply charging current from program 02 for utility charger.	40A (default) 	Setting range is 2A, then from 10A to 150A for HV10K model.
12	Setting voltage point back to utility source when selecting "SBU priority" or "Solar first" in program 01.	Available options:	
		46V (default) 	Setting range is from 44V to 51V. Increment of each click is 1V.
		Available options when any lithium battery type is selected in Program 05.	
		SOC 10% (default for Lithium) 	If any types of lithium battery is selected in program 05, setting value will change to SOC automatically. Adjustable range is 5% to 95%.
13	Setting voltage point back to battery mode when selecting "SBU priority" or "Solar first" in program 01.	Available options: Setting range is FUL and from 48V to 58V. Increment of each click is 1V.	
		Battery fully charged 	54V (default) 
		Available option when any lithium battery type is selected in Program 05.	
		SOC 80% (default for Lithium) 	If any types of lithium battery is selected in program 05, setting value will change to SOC automatically. Adjustable range is 10% to 100%. Increment of each click is 5%.

16	Charger source priority: To configure charger source priority	If this inverter/charger is working in Line, Standby or Fault mode, charger source can be programmed as below:	
		Solar first 16 CSO	Solar energy will charge battery as first priority. Utility will charge battery only when solar energy is not available.
		Solar and Utility (default) 16 SNU	Solar energy and utility will charge battery at the same time.
		Only Solar 16 OSO	Solar energy will be the only charger source no matter utility is available or not.
		If this inverter/charger is working in Battery mode, only solar energy can charge battery. Solar energy will charge battery if it's available and sufficient.	
18	Alarm control	Alarm on (default) 18 bON	Alarm off 18 bOF
19	Auto return to default display screen	Return to default display screen (default) 19 ESP	If selected, no matter how users switch display screen, it will automatically return to default display screen (Input voltage /output voltage) after no button is pressed for 1 minute.
		Stay at latest screen 19 KEP	If selected, the display screen will stay at latest screen user finally switches.
20	Backlight control	Backlight on (default) 20 LON	Backlight off 20 LOF
22	Beeps while primary source is interrupted	Alarm on (default) 22 AON	Alarm off 22 AOF
23	Overload bypass: When enabled, the unit will transfer to line mode if overload occurs in battery mode.	Bypass disable (default) 23 byd	Bypass enable 23 byE
25	Record Fault code	Record enable (default) 25 FEN	Record disable 25 FdS
26	Bulk charging voltage (C.V voltage)	default setting: 56.4V CV 26 BATT 56.4V	
		If self-defined is selected in program 5, this program can be set up. Setting range is from 48.0V to 61.0V. Increment of each click is 0.1V.	

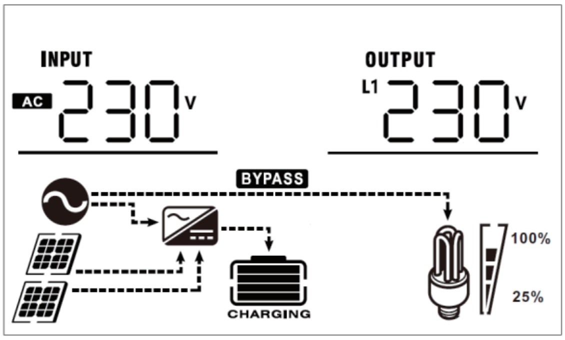
27	Floating charging voltage	default setting: 54.0V 	
		If self-defined is selected in program 5, this program can be set up. Setting range is from 48.0V to 61.0V. Increment of each click is 0.1V.	
28	Maximum discharging current of battery NOTE: The maximum discharging current of battery can be limited by rated power on battery mode. The unit will discharge with the maximum current without any protection when default "disable" is selected. User can set the value according to the actual battery spec. If discharging current is higher than setting value, battery will stop discharging.	Available options in HV10K models: Setting range is disable and from 30A to 230A. Increment of each click is 10A.	
		dIS (default) 	dIS (default) 
29	Low DC cut-off voltage or SOC percentage	default setting: 42.0V  If self-defined is selected in program 5, this program can be set up. Setting range is from 42.0V to 52.0V. Increment of each click is 0.1V. Low DC cut-off voltage will be fixed to setting value no matter what percentage of load is connected.	
		Lithium battery default setting: SOC 5%  If any type of lithium battery is selected in program 05, setting value will change to SOC automatically. Adjustable range is 0% to 90%. Increment of each click is 1%.	
30	Battery equalization	Battery equalization 	Battery equalization disable (default) 
		If "Flooded" or "User-Defined" is selected in program 05, this program can be set up.	
31	Battery equalization voltage	default setting: 58.4V 	
		Setting range is from 48.0V to 61.0V. Increment of each click is 0.1V.	
33	Battery equalized time	60min (default) 	Setting range is from 5min to 900min. Increment of each click is 5min.

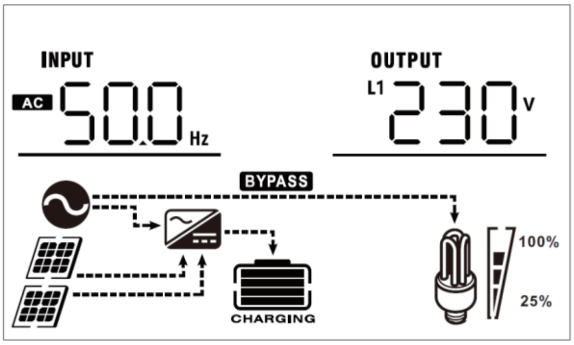
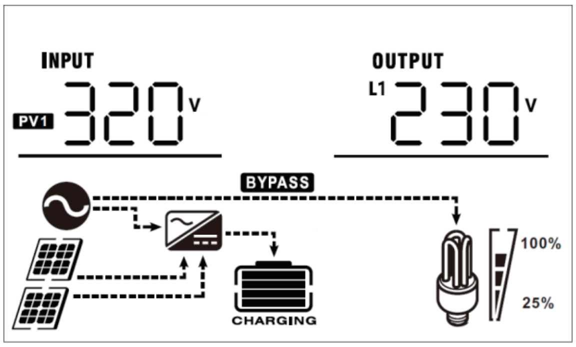
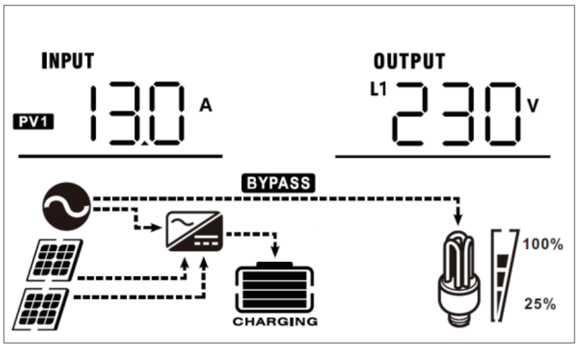
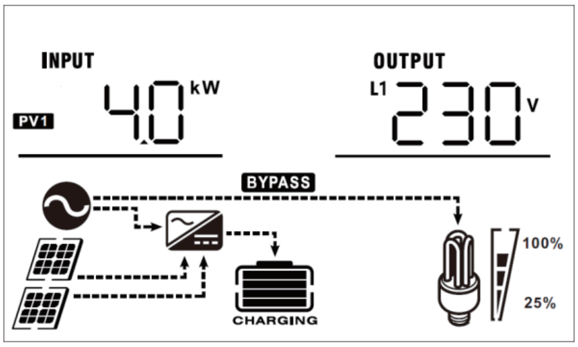
34	Battery equalized timeout	120min (default) 34 120	Setting range is from 5min to 900 min. Increment of each click is 5 min.
35	Equalization interval	30days (default) 35 30d	Setting range is from 0 to 90 days. Increment of each click is 1 day.
36	Equalization activated immediately	Enable 36 AEN	Disable (default) 36 AdS
		If equalization function is enabled in program 30, this program can be set up. If "Enable" is selected in this program, it's to activate battery equalization immediately and LCD main page will shows "E9". If "Disable" is selected, it will cancel equalization function until next activated equalization time arrives based on program 35 setting. At this time, "E9" will not be shown in LCD main page.	
42	AC input detection current *Note: To balance AC input current when an external device (like transformer, energy meter) connected at AC input.	When there is a deviation current caused by external devices connected at AC input, it can be balanced by adjusting the current. Setting range is from 0 to 250. Increment of each click is 1.	
		Nothing shown if unit is not in Line mode. 42	150 (default) will show if unit is in Line mode. 42 150
43	Power limit for PV energy in Line Mode *Note: This setting is to prevent excessive energy generated by PV exceeds load demand.	When there is a deviation of load detection caused by external devices connected at AC input, it can be adjusted by this setting. Setting range is from 0W to 250W. Increment of each click is 10W.	
		Nothing shown if unit is not in Line mode. 43	30W (default) will show if unit is in Line mode. 43 30
60	Low DC cut off voltage or SOC percentage on second output (L2)	default setting: 42.0V COU 60 BATT 42.0V If User-defined" is selected in program 05, this setting range is from 42.0 to 52.0V. Increment of each click is 0.1V. Low DC cut-off voltage will be fixed to setting value no matter what percentage of load is connected.	
		Lithium battery default setting: SOC 5% COU 60 BATT 5% If any types of lithium battery is selected in program 05, this program can be set up. Setting range is from 0% to 90%. Increment of each click is 1%.	
61	Setting discharge time on second output (L2)	Disable (default) 61 ddS	Setting range is disable and then from 0 min to 990 min. Increment of each click is 5 min. *If the battery discharge time achieves the setting time in Program 61 and the program 60 function is not triggered, the output will be turned off.

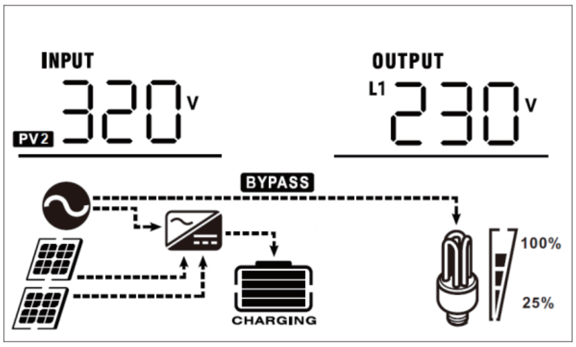
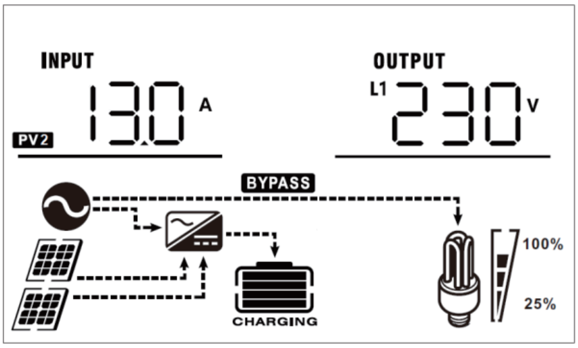
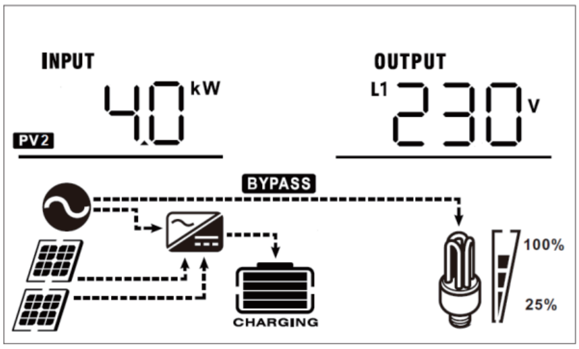
63	Setting voltage point or SOC to restart on the second output (L2)	default setting: 46.0V 	If “User-defined” is selected in program 05, this setting range is from 43.0V to 61.0V. Increment of each click is 0.1V. *If second output is cut off due to setting in program 60, second output (L2) will restart according to setting in program 63.
		SOC: 20% (default for lithium battery) 	If any type of lithium battery is selected in program 05, this parameter value will be displayed in percentage and value setting is based on battery capacity percentage. Setting range is from 5% to 100%. Increment of each click is 5%. *If second output is cut off due to setting in program 60, second output (L2) will restart according to setting in program 63.
64	Setting waiting time to turn on the second output when the inverter is back to Line mode or battery is in charging status	0 min (Default) 	Setting range is from 0 min to 990 min. Increment of each click is 5 min. *If second output is cut off due to setting in program 61, second output (L2) will restart according to setting in program 64.

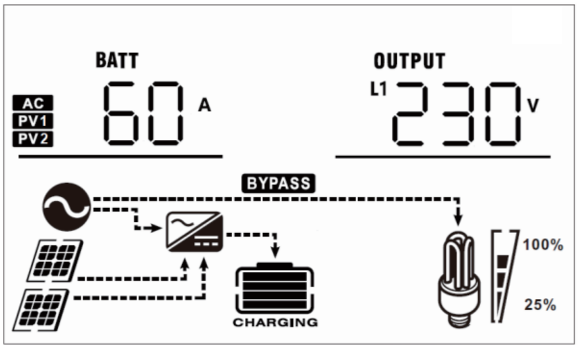
Display Setting

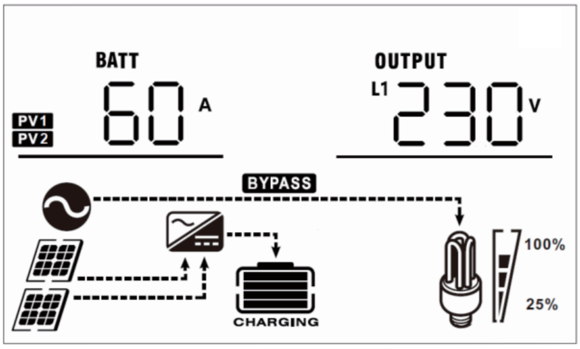
The LCD display information will be switched in turns by pressing “UP” or “DOWN” key. The selectable information is switched as following order in listed table.

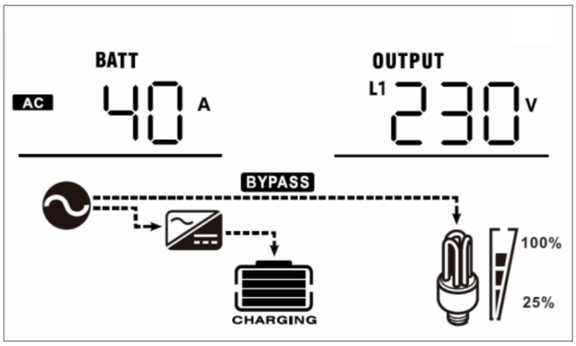
Selectable information	LCD display
Input voltage/Output voltage (Default Display Screen)	Input Voltage=230V, main output voltage=230V 

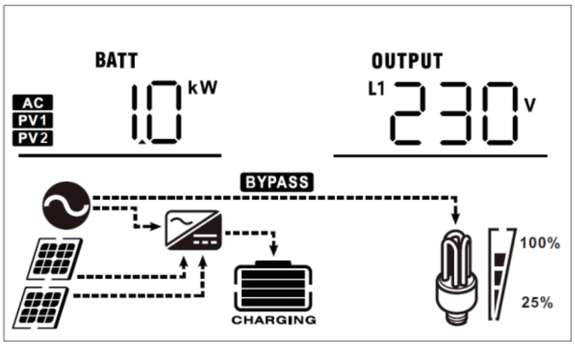
<table>
<tr><td>Input frequency</td><td>Input frequency=50Hz

</td></tr>
<tr><td>PV1 voltage</td><td>PV1 voltage=320V

</td></tr>
<tr><td>PV1 current</td><td>PV1 current = 13.0A

</td></tr>
<tr><td>PV1 power</td><td>PV1 power = 4.0kW

</td></tr>
</table>

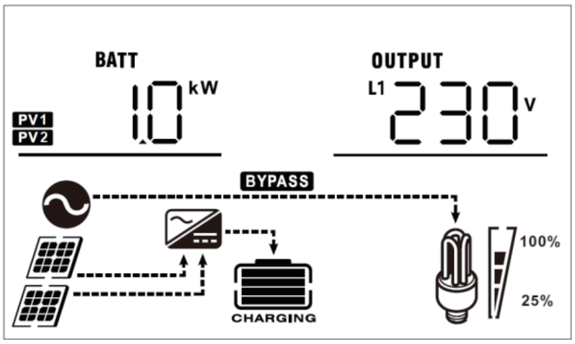
PV2 voltage	PV2 voltage=320V 
PV2 current	PV2 current = 13.0A 
PV2 power	PV2 power = 4.0kW 

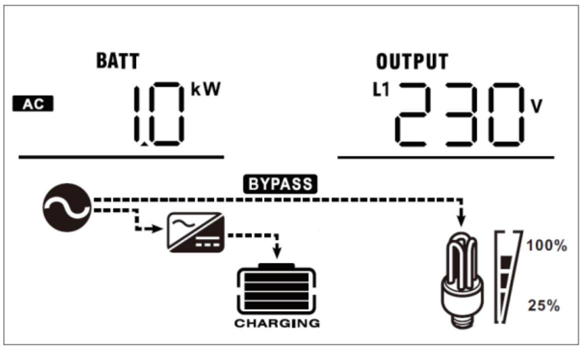
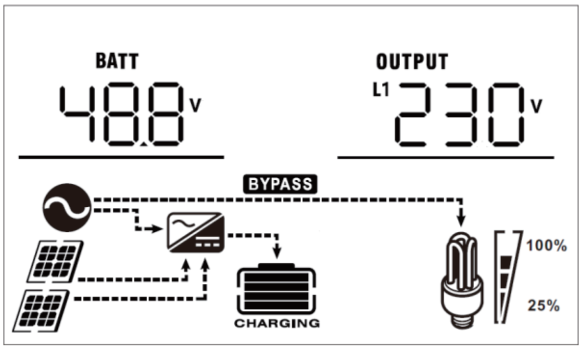
<table>
<tr><td>Charging current</td><td>AC and PV charging current=60A


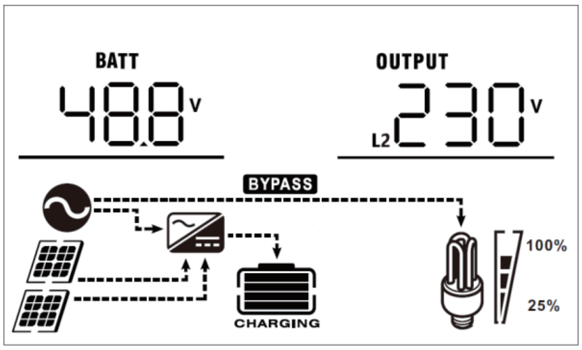
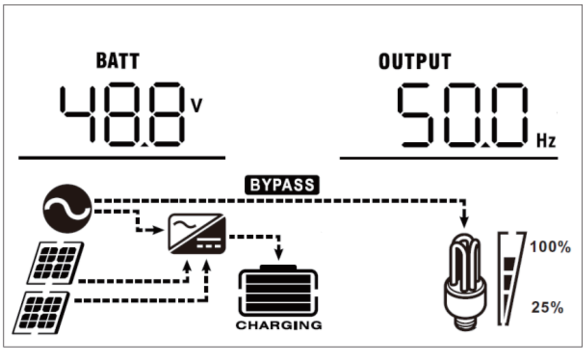
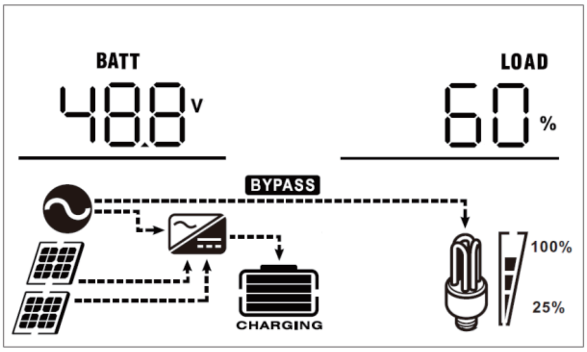
PV charging current=60A


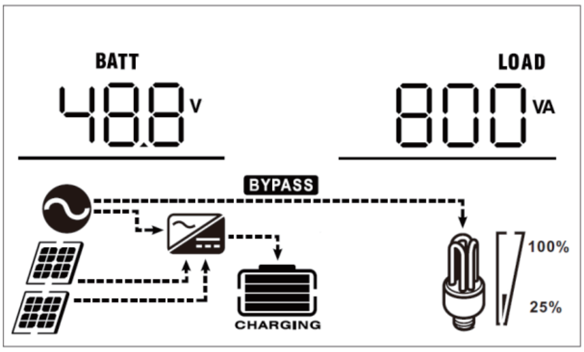
AC charging current=40A

</td></tr>
</table>

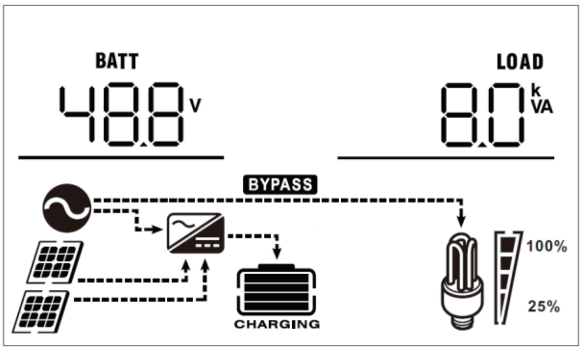
<table>
<tr>
<td>Charging power</td>
<td>AC and PV charging power=1.0kW


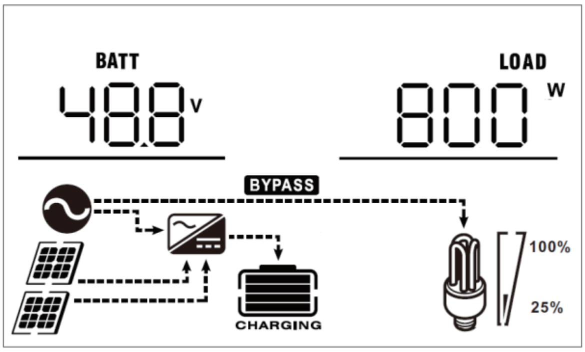
PV charging power=1.0kW


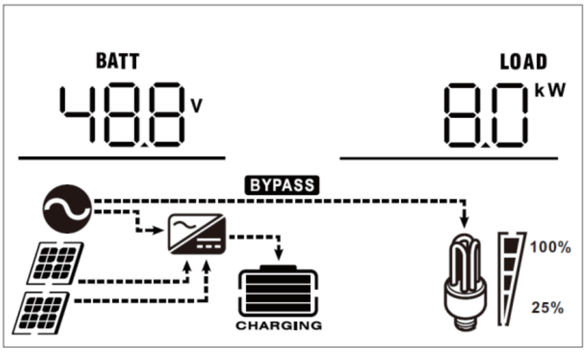
AC charging power=1.0kW

</td>
</tr>
<tr>
<td>Battery voltage and output voltage</td>
<td>Battery voltage=48.8V, main output voltage=230V

</td>
</tr>
</table>

Battery voltage and second output voltage	Battery voltage=48.8V, second output voltage=230V 
Output frequency	Output frequency=50Hz 
Load percentage	Load percent=60% 

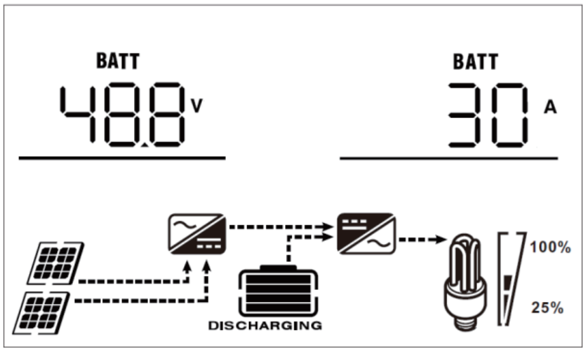
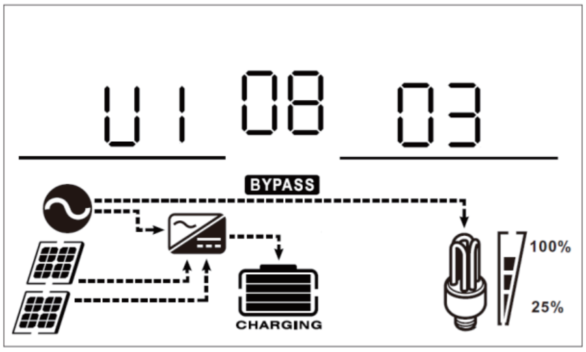
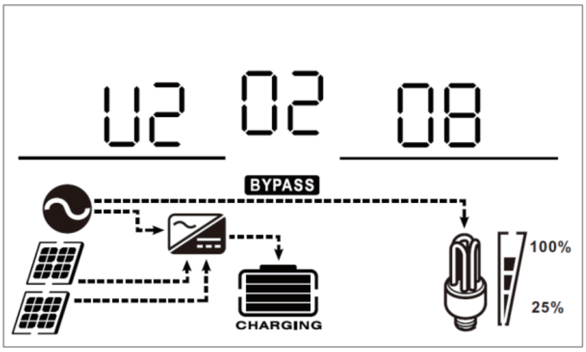
<table>
<tr>
<td>Load in VA</td>
<td>When connected load is lower than 1kVA, load in VA will present xxxVA like below chart.


When load is larger than 1kVA (≧1KVA), load in VA will present x.xkVA like below chart.


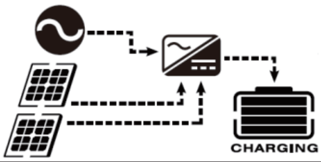
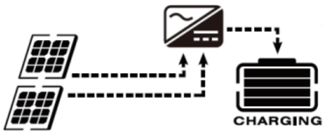
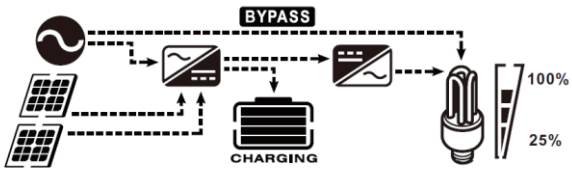
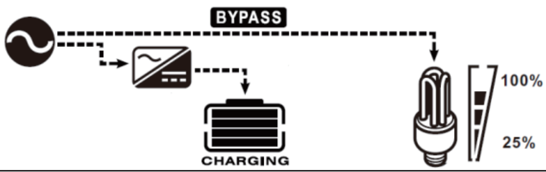
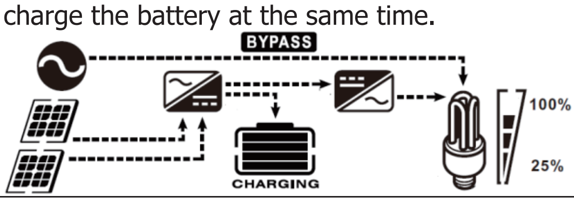
</td>
</tr>
<tr>
<td>Load in Watt</td>
<td>When load is lower than 1kW, load in W will present xxxW like below chart.


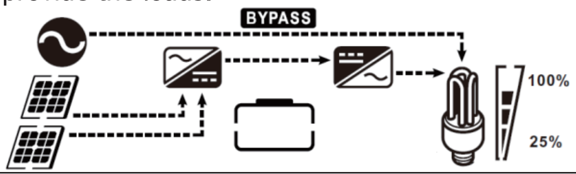
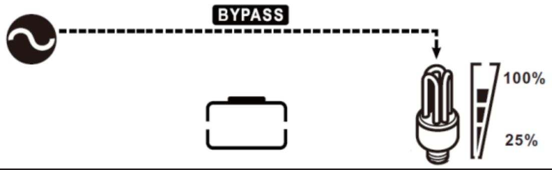
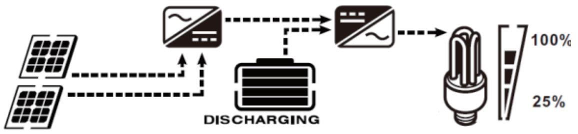
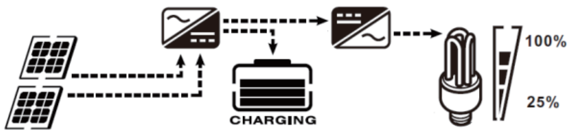
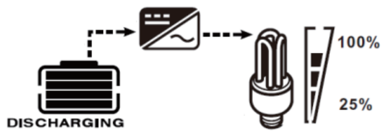
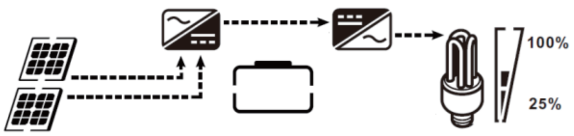
When load is larger than 1kW (≧1KW), load in W will present x.xkW like below chart.


</td>
</tr>
</table>

Battery voltage/DC discharging current	Battery voltage=48.8V, discharging current=30A 
Main CPU version checking	Main CPU version 00008.03 
Secondary CPU version checking	Secondary CPU version 00002.08 

Operating Mode Description

Operation mode	Description	LCD display
Standby mode **Note:** *Standby mode: The inverter is not turned on yet but at this time, the inverter can charge battery without AC output.	No output is supplied by the unit but it still can charge batteries.	Charging by utility and PV energy. 
		Charging by utility. 
		Charging by PV energy. 
		No charging. 
Fault mode **Note:** *Fault mode: Errors are caused by inside circuit error or external reasons such as over temperature, output short circuited and so on.	PV energy and utility cannot charge batteries.	No charging, no output. 
Line Mode	The unit will provide output power from the mains. It will also charge the battery at line mode.	Charging by utility and PV energy. 
		Charging by utility. 
		If "solar first" is selected as output source priority and solar energy is not sufficient to provide the load, solar energy and the utility will provide the loads and charge the battery at the same time. 

<table>
<tr><th>Operation mode</th><th>Description</th><th>LCD display</th></tr>
<tr><td rowspan="2">Line Mode</td><td rowspan="2">The unit will provide output power from the mains. It will also charge the battery at line mode.</td><td>If “solar first” is selected as output source priority and battery is not connected, solar energy and the utility will provide the loads.

</td></tr>
<tr><td>Power from utility.

</td></tr>
<tr><td rowspan="4">Battery Mode</td><td rowspan="4">The unit will provide output power from battery and PV power.</td><td>Power from battery and PV energy.

</td></tr>
<tr><td>PV energy will supply power to the loads and charge battery at the same time.

</td></tr>
<tr><td>Power from battery only.

</td></tr>
<tr><td>Power from PV energy only.

</td></tr>
</table>

Battery Equalization Description

Equalization function is added into charge controller. It reverses the buildup of negative chemical effects like stratification, a condition where acid concentration is greater at the bottom of the battery than at the top. Equalization also helps to remove sulfate crystals that might have built up on the plates. If left unchecked, this condition, called sulfation, will reduce the overall capacity of the battery. Therefore, it’s recommended to equalize battery periodically.

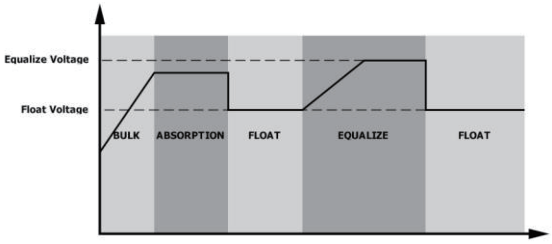
- **How to Apply Equalization Function**

You must enable battery equalization function in monitoring LCD setting program 30 first. Then, you may apply this function in device by either one of following methods:

1. Setting equalization interval in program 35.
2. Active equalization immediately in program 36.

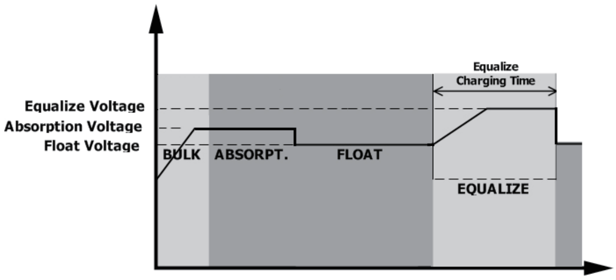
- **When to Equalize**

In float stage, when the setting equalization interval (battery equalization cycle) is arrived, or equalization is active immediately, the controller will start to enter Equalize stage.

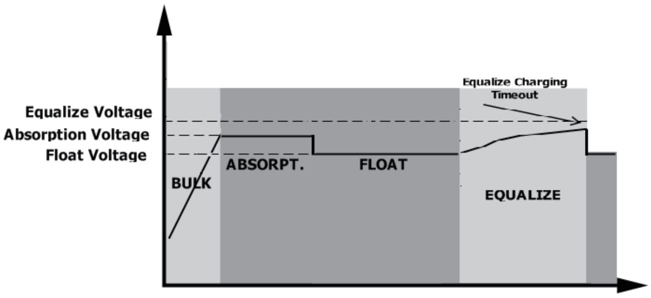


- **Equalize charging time and timeout**

In Equalize stage, the controller will supply power to charge battery as much as possible until battery voltage raises to battery equalization voltage. Then, constant-voltage regulation is applied to maintain battery voltage at the battery equalization voltage. The battery will remain in the Equalize stage until setting battery equalized time is arrived.



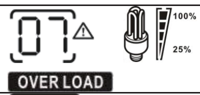
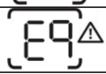
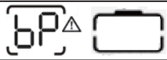
However, in Equalize stage, when battery equalized time is expired and battery voltage doesn't rise to battery equalization voltage point, the charge controller will extend the battery equalized time until battery voltage achieves battery equalization voltage. If battery voltage is still lower than battery equalization voltage when battery equalized timeout setting is over, the charge controller will stop equalization and return to float stage.



Fault Reference Code

Fault Code	Fault Event	Icon on
01	Fan is locked when inverter is off.	01 ERROR
02	Over temperature or NTC is not connected well.	02 ERROR
03	Battery voltage is too high	03 ERROR
04	Battery voltage is too low	04 ERROR
05	Output short circuited or over temperature is detected by internal converter components.	05 ERROR
06	Output voltage is too high.	06 ERROR
07	Overload time out	07 ERROR
08	Bus voltage is too high	08 ERROR
09	Bus soft start failed	09 ERROR
51	Over current or surge	51 ERROR
52	Bus voltage is too low	52 ERROR
53	Inverter soft start failed	53 ERROR
55	Over DC voltage in AC output	55 ERROR
57	Current sensor failed	57 ERROR
58	Output voltage is too low	58 ERROR
59	PV voltage is over limitation	59 ERROR

Warning Indicator

Warning Code	Warning Event	Audible Alarm	Icon flashing
01	Fan is locked when inverter is on.	Beep three times every second	
02	Over temperature	None	
03	Battery is over-charged	Beep once every second	
04	Low battery	Beep once every second	
07	Overload	Beep once every 0.5 second	
10	Output power derating	Beep twice every 3 seconds	
15	PV energy is low.	Beep twice every 3 seconds	
16	High AC input (>280VAC) during BUS soft start	None	
32	Communication failure between inverter and communication board	None	
E9	Battery equalization	None	
bP	Battery is not connected	None	

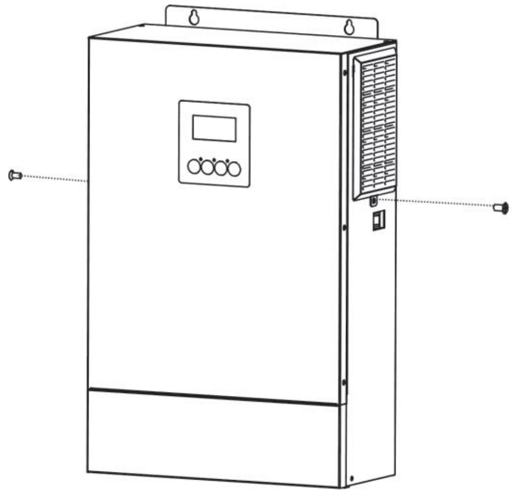
CLEARANCE AND MAINTENANCE FOR ANTI-DUST KIT

Overview

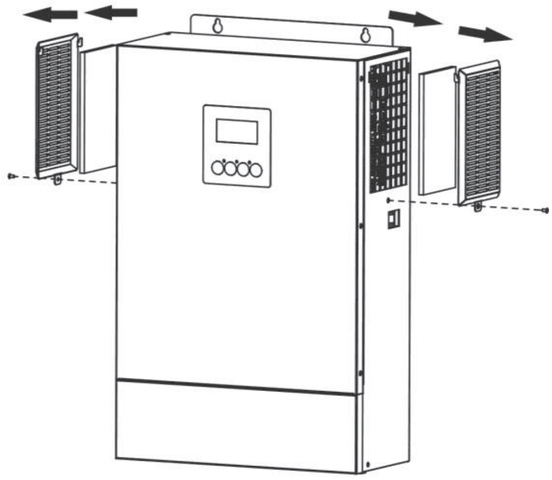
Every inverter is already installed with anti-dust kit from factory. This kit keeps dust from your inverter and increases product reliability in harsh environment.

Clearance and Maintenance

Step 1: Please loosen the screw in counterclockwise direction on the top or both sides of the inverter.



Step 2: Then, dustproof case can be removed and take out air filter foam as shown in below chart.



Step 3: Clean air filter foam and dustproof case. After clearance, re-assemble the dust-kit back to the inverter.

NOTICE: The anti-dust kit should be cleaned from dust every one month.

SPECIFICATIONS

Table 1 Line Mode Specifications

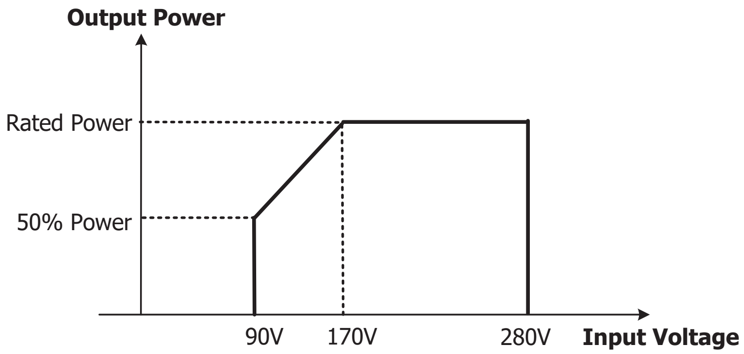
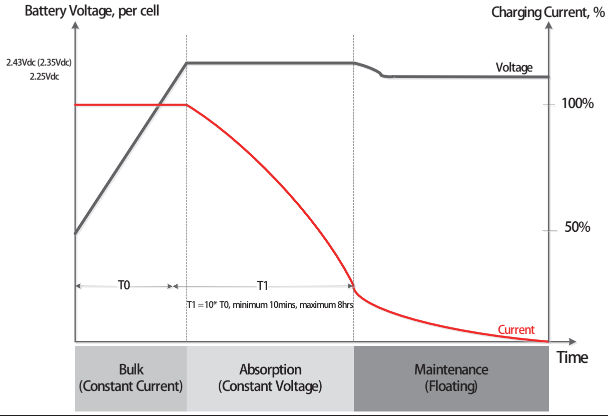
INVERTER MODEL	HV10K
Input Voltage Waveform	Sinusoidal (utility or generator)
Nominal Input Voltage	230Vac
Low Loss Voltage	170Vac±7V (UPS); 90Vac±7V (Appliances)
Low Loss Return Voltage	180Vac±7V (UPS); 100Vac±7V (Appliances)
High Loss Voltage	280Vac±7V
High Loss Return Voltage	270Vac±7V
Max AC Input Voltage	300Vac
Nominal Input Frequency	50Hz / 60Hz (Auto detection)
Low Loss Frequency	40±1Hz
Low Loss Return Frequency	42±1Hz
High Loss Frequency	65±1Hz
High Loss Return Frequency	63±1Hz
Output Short Circuit Protection	Circuit Breaker
Efficiency (Line Mode)	>95% (Rated R load, battery full charged)
Transfer Time	10ms typical (UPS); 20ms typical (Appliances)
Output power derating: When AC input voltage drops to 170V, the output power will be derated.	

Table 2 Inverter Mode Specifications

INVERTER MODEL	HV10K
Rated Output Power	HV10KVA/HV10KW
Output Voltage Waveform	Pure Sine Wave
Output Voltage Regulation	230Vac±5%
Output Frequency	50Hz
Peak Efficiency	93%
Overload Protection	5s@≥120% load; 30s@103%~120% load
Surge Capacity	2* rated power for 5 seconds
Nominal DC Input Voltage	48Vdc
Cold Start Voltage	46.0Vdc
Low DC Warning Voltage @ load < 50% @ load ≥ 50%	 46.0Vdc 44.0Vdc
Low DC Warning Return Voltage @ load < 50% @ load ≥ 50%	 47.0Vdc 46.0Vdc
Low DC Cut-off Voltage @ load < 50% @ load ≥ 50%	 43.0Vdc 42.0Vdc
High DC Recovery Voltage	62Vdc
High DC Cut-off Voltage	63Vdc
No Load Power Consumption	78W
AC output current	43.5A
AC input surge current and duration	87A, 5s
DC input/output current	230A
Battery nominal input/ output voltage	48V

Table 3 Charge Mode Specifications

Utility Charging Mode		
INVERTER MODEL		**HV10K**
Charging Algorithm		3-Step
AC Charging Current (Max)		150Amp (@$V_{I/P}$=230Vac)
Bulk Charging Voltage	**Flooded Battery**	58.4Vdc
	AGM / Gel Battery	56.4Vdc
Floating Charging Voltage		54Vdc
Charging Curve		

MPPT Solar Charging Mode	
INVERTER MODEL	**HV10K**
Max. PV Array Power	14000W (7000W x 2)
Nominal PV Voltage	320Vdc
Start-up Voltage	150Vdc +/- 10Vdc
PV Array MPPT Voltage Range	90~450Vdc (Min. 100V without Battery)
Max. PV Array Open Circuit Voltage	500Vdc
Max. Input Current	22Amp x 2
Max Charging Current **(AC charger plus solar charger)**	180Amp

Table 4 General Specifications

INVERTER MODEL	HV10K
Safety Certification	CE
Operating Temperature Range	-10°C to 50°C
Storage temperature	-15°C~ 60°C
Humidity	5% to 95% Relative Humidity (Non-condensing)
Dimension (D*W*H), mm	141 x 322 x 497
Net Weight, kg	16.3
Electric shock protection rating	Class I
IP rating	IP21
Overvoltage rating for each port	Mains (oV/C ll), PV (oV/C ll)
Maximum operating altitude	0-1500 meters
Environmental pollution level	PD2
Warranty*	2 Years

*This Warranty does not apply to any defect or failure of performance that includes or is caused by any of the circumstances listed in **Appendix I.**

TROUBLE SHOOTING

Problem	LCD/LED/Buzzer	Explanation / Possible cause	What to do
Unit shuts down automatically during startup process.	LCD/LEDs and buzzer will be active for 3 seconds and then complete off.	The battery voltage is too low	1. Re-charge battery. 2. Replace battery.
No response after power on.	No indication.	1. The battery voltage is far too low. 2. Internal fuse tripped.	1. Contact repair center for replacing the fuse. 2. Re-charge battery. 3. Replace battery.
Mains exist but the unit works in battery mode.	Input voltage is displayed as 0 on the LCD and green LED is flashing.	Input protector is tripped	Check if AC breaker is tripped and AC wiring is connected well.
	Green LED is flashing.	Insufficient quality of AC power. (Shore or Generator)	1. Check if AC wires are too thin and/or too long. 2. Check if generator (if applied) is working well or if input voltage range setting is correct. (UPS→Appliance)
	Green LED is flashing.	Set "Solar First" as the priority of output source.	Change output source priority to Utility first.
When the unit is turned on, internal relay is switched on and off repeatedly.	LCD display and LEDs are flashing	Battery is disconnected.	Check if battery wires are connected well.
Buzzer beeps continuously and red LED is on.	Fault code 07	Overload error. The inverter is overload 105% and time is up.	Reduce the connected load by switching off some equipment.
		If PV input voltage is higher than specification, the output power will be derated. At this time, if connected loads is higher than derated output power, it will cause overload.	Reduce the number of PV modules in series or the connected load.
	Fault code 05	Output short circuited.	Check if wiring is connected well and remove abnormal load.
	Fault code 02	Internal temperature of inverter component is over.	Check whether the air flow of the unit is blocked or whether the ambient temperature is too high.
	Fault code 03	Battery is over-charged.	Return to repair center.
		The battery voltage is too high.	Check if spec and quantity of batteries are meet requirements.
	Fault code 01	Fan fault	Replace the fan.
	Fault code 06/58	Output abnormal (Inverter voltage is too low or too high)	1. Reduce the connected load. 2. Return to repair center
	Fault code 08/09/53/57	Internal components failed.	Return to repair center.
	Fault code 51	Over current or surge.	Restart the unit, if the error happens again, please return to repair center.
	Fault code 52	Bus voltage is too low.	
	Fault code 55	Output voltage is unbalanced.	
	Fault code 59	PV input voltage is beyond the specification.	Reduce the number of PV modules in series.

Appendix I.

This Warranty does not apply to any defect or failure of performance that includes or is caused by any of the following.

1) The Product is not installed and/or repaired by BLUETTI's certified installers,

2) The Warranty Period for the Product has lapsed due to time or aggregate energy throughput limitation

3) The Buyer fails to notify BLUETTI or a BLUETTI authorized service partner ("BLUETTI Partner" or "BLUETTI Partners") of the defect or deterioration within 30 days of the Buyer becoming aware of the defect or deterioration.

4) Failure to comply with product user manuals, maintenance guidelines and other products on the use of the Product.

5) Shipping damage (including scratches caused by movement of the inner pack aging during shipping). In this case claims shall be raised directly to the shipping company or insurance company as soon as possible and obtain identification of damage such as container/package unloading; malfunctions caused by the shipping process.

6) Failure of or damage to the machine caused by installation, repair, alteration or disassembly without BLUETTI and/or not by BLUETTI's authorized personnel.

7) If the Product has been out of operation for 6 months or more.

8) If the Product is not manufactured by BLUETTI.

9) If the Product is not transported, stored, installed, assembled, fastened or wired in accordance with the official installation manuals and data sheets of the Product.

10) Defects or performance failures of the Products due to misuse, intentional mis conduct or negligence of the Buyer.

11) Buyer uses the Product in violation of applicable laws and safety regulations,in cluding national, state, or local government laws or regulations.

12) Damage caused by abnormal voltage (such as excessive voltage, abnormal power failure) or computer virus infection.

13) The Product is damaged due to improper storage. Failures and damages caused by unforeseen or human factors or force majeure, such as storms, earthquakes, floods, lightning, overpressure, pests, fires, theft and weak signals causing monitoring failures.

14) The identification information (trademark, barcode, model) on the Product is not clear, deliberately damaged or stained, or indelible marks (such as paint) are made on the identification information.

15) The Products are used or installed in places where vibration is expected to occur.

16) Failure of the Product caused by the connection to incompatible test equipment (such as DC (direct current) analog power supply).

17) Product failures not caused by the quality of the BLUETTI Product itself, including but not limited to failures caused by the use of non-BLUETTI original parts, failures caused by third party software or hardware incompatibility, etc.

18) Products authorized for trial, testing, training or demonstration purposes only.

19) Products purchased from sources not authorized by BLUETTI and Products purchased from overseas sales channels shall not be covered by the warranty in the home countries of the Buyer.

20) Products used to power medical equipment or life support equipment.

21) The Products used in facilities with radiation control, nuclear reactors, and facilities related to nuclear safety or nuclear power uses, as well as facilities that may have direct contact with patients.

22) The Product is applied to non-residential fixed storage applications;

23) In outdoor installation, Product is exposed to direct sunlight and rain.Because battery must not be mounted in direct sunlight as this affects the internal heat, and therefore efficiency of the battery and warranty.

24) The maximum temperature of the environment where the Products are located is above 50 °C for more than 5 hours at a time.

25) The minimum temperature of the environment where the Products are located is below -10 °C for more than 7 days at a time or below -20 °C for more than 24 hours at a time.

Need Help? We're here for you!

Customer Service

Email: sale-nga@bluettipower.com

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BLUETTI

Certificate

Inspector: ____________________

QC: ____________________

Always Share Excellence